

Tendencies: PROE, Personnel, Motion, and Coverage from Charting Data

Separating what a team chooses from what the scoreboard forces, and turning a season of charted snaps into a one-page card

In 2025 the Los Angeles Rams dropped back to pass on 59% of their snaps and the New York Jets on 61%. By that number, the one a broadcast graphic would show you, they were the same kind of offense. They were close to opposites. The Rams, with Matthew Stafford on his way to the MVP award and a league-leading 46 touchdown passes,¹ threw when they didn't have to. The Jets threw because they were behind: they trailed on three snaps in four. Whenever a Jets game was close and the clock wasn't a factor, they *ran* more than any other offense in football. "The Jets threw the ball 61% of the time" describes the Jets' scoreboard, not their play-caller. This chapter is about measuring the choice instead of the circumstance: what a team does when it could do anything, how to compute it from public data, how far to trust it, and how to fold it, with the personnel, motion and coverage charting that sits beside it, into a one-page card you can take to a game.

You'll be able to:

- Explain why a raw pass rate mostly measures the score, and pick out the **neutral situations** where play-calling tendencies show.
- Explain expected pass rate (**xpass**) and pass rate over expected (**PROE**), fit a small xpass model yourself, and compute PROE by team, season and situation.
- Measure an offense's personnel, quarterback-alignment, motion, play-action and no-huddle tendencies, and read them side by side in a league dashboard.
- Measure a defense's coverage tendencies from charted labels, and say what those labels can and cannot tell you.
- State the caveats of charting data (definitions, judgement and error, the 2022/2023 source break, missing seasons) before you quote a number.
- Build a one-page data scouting report on an opponent, and grade it live during the game.

i You'll need

- **Expected Points, EPA, and Success Rate:** EPA (expected points added, the change in a situation's point value on one play) as the yardstick for "did it work?", and the habit of asking whether a number is signal or noise.
- **Constraint Theory and Sequencing:** a **tendency** (a pattern an opponent can find and play against), the **tendency breaker** and the **self-scout**, and the hand-built tendency table of pass rate by personnel and quarterback alignment. This chapter turns that table into league-wide measures.
- **Working with nflverse Data in Python:** loading play-by-play with nflreadpy, and the two charting files joined to it, participation data and FTN charting data.

The data this chapter uses

Three public files, joined on game and play, carry everything in this chapter. Like the rest of Part 13, the code is shown, so read the cells as part of the text. `gridiron.data` wraps nflreadpy and caches each season under `data/cache/`.

```
import re
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.colors import LinearSegmentedColormap
from IPython.display import Markdown, display
from gridiron import style
from gridiron.data import pbp, nfl
```

¹AP awards, 2025 season (NFL Honors, February 5, 2026): Matthew Stafford MVP, NFL-leading 46 touchdown passes. <https://amp.nfl.com/news/list-of-nfl-honors-award-winners-from-2025-nfl-season>. Pass rates, trailing shares and win probabilities in this chapter are its own calculations from nflverse play-by-play (regular season; runs and dropbacks, sacks and scrambles counted as dropbacks).

```

style.apply()
SEASONS = range(2016, 2026)          # ten completed seasons, 2016-2025
COLS = ["season", "season_type", "week", "game_id", "play_id", "posteam", "defteam", "play_type",
        "pass", "qb_dropback", "down", "ydstogo", "yardline_100", "qtr", "half_seconds_remaining",
        "game_seconds_remaining", "score_differential", "wp", "xpass", "pass_oe", "epa", "success"]
raw = pbp(SEASONS, columns=COLS)

# Scrimmage snaps: designed runs and dropbacks (sacks and scrambles count as dropbacks).
# Kneels, spikes, kicks and snaps wiped out by a penalty are left out.
plays = raw[(raw.season_type == "REG") & raw.play_type.isin(["pass", "run"])
            & raw.down.notna() & raw.wp.notna()].copy()

def is_neutral(d, early_downs=True):
    """The course's neutral situation: win probability 20-80% and more than two minutes left in
    the half. For play-calling tendencies, first and second down only."""
    m = d.wp.between(0.20, 0.80) & (d.half_seconds_remaining > 120)
    return m & (d.down <= 2) if early_downs else m

plays["neutral"] = is_neutral(plays, early_downs=False)
plays["early_neutral"] = is_neutral(plays)

# Charting: personnel, box count, rushers and coverage (participation, 2018-2025) and
# quarterback alignment, motion, play-action, no-huddle and blitzers (FTN, 2022-2025).
part = (nfl().load_participation(list(range(2018, 2026))).to_pandas()
        .rename(columns={"nflverse_game_id": "game_id"})
        .[[ "game_id", "play_id", "offense_personnel", "defense_personnel", "defenders_in_box",
            "number_of_pass_rushers", "defense_man_zone_type", "defense_coverage_type"]])
ftn = (nfl().load_ftn_charting([2022, 2023, 2024, 2025]).to_pandas()
        .rename(columns={"nflverse_game_id": "game_id", "nflverse_play_id": "play_id"})
        .[[ "game_id", "play_id", "qb_location", "is_motion", "is_play_action", "is_no_huddle", "n_blitzers"]])
snaps = (plays[plays.season >= 2018]
        .merge(part, on=["game_id", "play_id"], how="left")
        .merge(ftn, on=["game_id", "play_id"], how="left"))
assert len(snaps) == (plays.season >= 2018).sum()          # no duplicated snaps from the joins
for c in ["is_motion", "is_play_action", "is_no_huddle"]:
    snaps[c] = snaps[c].astype(float)                    # True/False/missing -> 1/0/NaN

def personnel_code(s):
    """'1 C, 2 G, 1 QB, 1 RB, 2 T, 1 TE, 3 WR' (2023 on) or '1 RB, 1 TE, 3 WR' (before) -> '11'."""
    if not isinstance(s, str):
        return None
    n = {k: int(v) for v, k in re.findall(r"(\d+) ([A-Z]+)", s)}
    return f"{n.get('RB', 0) + n.get('FB', 0)}{n.get('TE', 0)}"

snaps["pers"] = snaps.offense_personnel.map(personnel_code)
snaps["heavy"] = snaps.pers.map(lambda p: float(int(p[0]) >= 2 or int(p[1]) >= 3)
                                if isinstance(p, str) else np.nan) # 2+ backs or 3+ tight ends
snaps["uc"] = (snaps.qb_location == "U").astype(float).where(snaps.qb_location.notna())

# Coverage: labelled dropbacks only ("blown" and "combination" tags, and prevent, set aside).

```

```

TWO_HIGH = ["COVER_2", "2_MAN", "COVER_4", "COVER_6"]
cov = snaps[(snaps.qb_dropback == 1) & snaps.defense_coverage_type.notna()
            & ~snaps.defense_coverage_type.isin(["BLOWN", "COMBO", "PREVENT"])].copy()
cov["man"] = (cov.defense_man_zone_type == "MAN_COVERAGE").astype(float)
cov["two_high"] = cov.defense_coverage_type.isin(TWO_HIGH).astype(float)

def pct(x, nd=0):
    return f"{100 * x:.{nd}f}%"

def ordinal(n):
    n = int(n)
    return f"{n}{'th' if 10 <= n % 100 <= 20 else {1: 'st', 2: 'nd', 3: 'rd'}.get(n % 10, 'th')}}"

def md_table(df, digits=2):
    """Show a small DataFrame as a Markdown table (presentation plumbing, not football)."""
    out = df.copy()
    for c in out.select_dtypes("number"):
        whole = (out[c].dropna() % 1 == 0).all()
        out[c] = out[c].map(lambda v: f"{v:.,}{0 if whole else digits}f")
    align = ["---:" if pd.api.types.is_numeric_dtype(df[c]) else ":-:" for c in df]
    rows = [list(out.columns), align] + out.astype(str).values.tolist()
    return Markdown("\n".join(" | " + " | ".join(map(str, r)) + " | " for r in rows))

```

Three choices in that cell carry through the chapter. **Regular season only**, as in 13-02. **“Pass” means a dropback**: nflverse’s `pass` flag counts sacks and scrambles, because they began as pass calls, and that is the event an expected-pass model predicts. And **personnel is backs then tight ends** (“11” is one back and one tight end), the way Personnel Groupings taught it, rebuilt from the participation file’s position counts because that file wrote personnel one way through 2022 and another way from 2023.

One snap, as the data sees it

Before any modelling, it helps to see what the files actually say about a single play. Figure 1 is a typical first-and-10, drawn the way this book draws every snap, with the columns that describe it written on the picture.

```

from gridiron import Play, Player, offense, defense, routes as r, show_animation

def say(fld, play, text, at, to=None, ha="center", fs=7.2):
    """A boxed label at (d, w), with an optional thin arrow to the point `to`=(d, w)."""
    P = lambda d, w: tuple(float(v) for v in fld.xy(*play.to_abs(d, w))) # noqa: E731
    kw = dict(fontsize=fs, color=style.INK, ha=ha, va="center", zorder=12, linespacing=1.2,
              bbox=dict(boxstyle="round,pad=0.3", fc=style.SURFACE, ec=style.BASELINE, lw=0.6))
    if to is None:
        fld.ax.text(*P(*at), text, **kw)
    else:
        fld.ax.annotate(text, xy=P(*to), xytext=P(*at), **kw,
                        arrowprops=dict(arrowstyle="-", color=style.INK_2, lw=0.7, shrinkA=0, shrinkB=4))

off = offense("ace") # 12 personnel, under center
base = defense("4-3_over", "single_high", off)
snap = Play(off, base, los=45, ball_y="left", to_go=10, title="1st & 10, own 35")
snap.motion("Z", to=(-1.5, 8.6)) # in toward the formation, then set
fld = snap.draw(lateral=(-21, 21), window=(-12.8, 17.5), width_in=6.4)

```

```
say(fld, snap, "offense_personnel\n1 RB, 2 TE, 2 WR = \"12\\\"\", at=(-9.6, -11.0), to=(-7.0, -0.9))
say(fld, snap, "qb_location = U\n(under center)", at=(-4.6, -11.5), to=(-1.9, -0.8))
say(fld, snap, "is_motion = TRUE\n(Z moved, then set)", at=(-5.0, 14.8), to=(-1.5, 12.0))
say(fld, snap, "defenders_in_box: 7 or 8?\nis this safety in the box?", at=(11.8, 12.6), to=(7.6, 6.4))
say(fld, snap, "defense_personnel\n4 DL, 3 LB, 4 DB (base)", at=(15.4, -10.0))
say(fld, snap, "down 1, ydstogo 10, yardline_100 65\n(play-by-play adds score, clock, wp)",
    at=(-9.9, 9.6), fs=6.6)
plt.show()
```

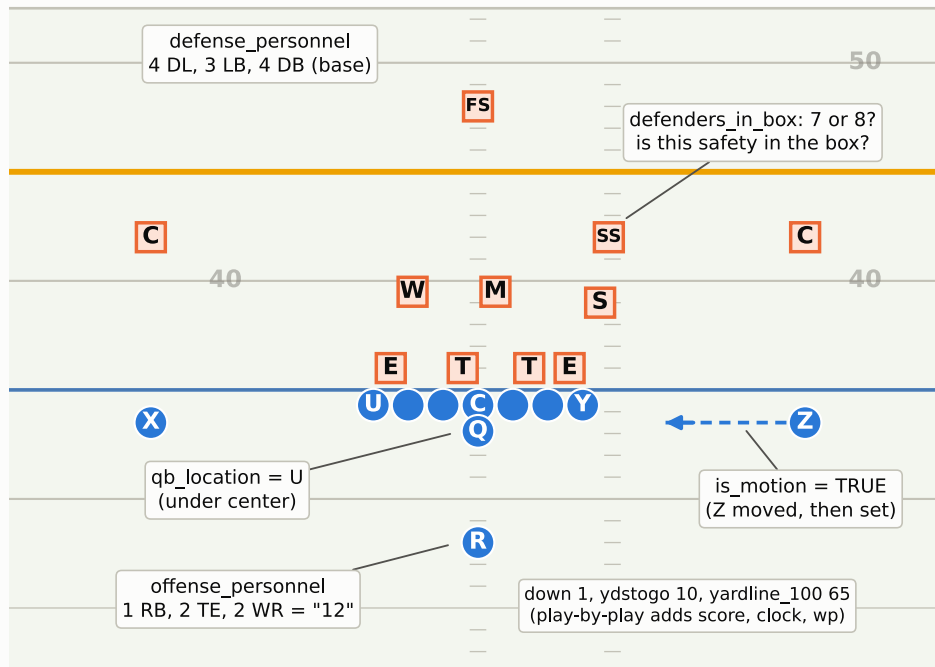


Figure 1: One snap and the columns that describe it. First-and-10 at the offense’s own 35, ball on the left hash: 12 personnel (one back, Y and U tight ends, X and Z receivers) with the quarterback under center, against a base 4-3 with one deep safety. Z motions in toward the formation and sets before the snap. Before the snap the files record the personnel (participation), where the quarterback stands and whether anyone moved (FTN), and the box count, which takes a judgement: is the strong safety, 7 yards deep and just outside the tight end, in the box or not? After the snap they add the play’s result and EPA (play-by-play), whether it was play-action and how many blitzed (FTN), and the number of pass rushers and the coverage label (participation).

File (nflreadpy loader)	What it adds to a snap	Seasons	When it arrives
Play-by-play (<code>load_pbp</code>)	Down, distance, field position, score, clock, result, EPA, win probability, nflverse's <code>xpass</code> and <code>pass_oe</code>	1999 on (<code>xpass</code> from 2006)	During the season, within hours of each game
Participation (<code>load_participation</code>)	Personnel on both sides, box count, number of pass rushers, coverage label and man/zone	2016 on; coverage 2018 on	After the season ends (2023 on)
FTN charting (<code>load_ftn_charting</code>)	Quarterback alignment, motion, play-action, no-huddle, RPO, screen, blitzers, and more	2022 on	During the season, as FTN charts each game

The last column matters more than it looks. As this chapter is written, in October 2026, the 2026 play-by-play and FTN files are filling in week by week, but 2026 *personnel and coverage* labels won't exist until after the season.² A scouting report on a 2026 opponent has to borrow those from 2025, or chart them yourself.

Why a raw pass rate lies

Start with the number everyone quotes. Figure 2 plots each 2025 offense's share of dropbacks across *all* its snaps against how well its games were going: the average win probability on its snaps (the model's chance that the team with the ball wins from there; Win Probability and Decision Analysis builds it).

```
RAW = plays.groupby(["season", "posteam"]).agg(
    raw=("pass", "mean"), wp=("wp", "mean"), epa_all=("epa", "mean"),
    trail=("score_differential", lambda s: (s < 0).mean()))
RAW25 = RAW.loc[2025]
R_RAW_WP = RAW25.raw.corr(RAW25.wp)
# the opening paragraph quotes these
assert round(100 * RAW25.loc["LA", "raw"]) == 59 and round(100 * RAW25.loc["NYJ", "raw"]) == 61
assert 0.72 < RAW25.loc["NYJ", "trail"] < 0.78

fig, ax = plt.subplots(figsize=(6.4, 3.7))
x, y = RAW25.wp * 100, RAW25.raw * 100
b = np.polyfit(x, y, 1)
xs = np.linspace(x.min() - 1, x.max() + 1, 20)
ax.plot(xs, np.polyval(b, xs), color=style.BASELINE, lw=1.3, zorder=1)
ax.scatter(x, y, s=22, color=style.MUTED, alpha=0.75, zorder=2)
for tm, col, txt, dx, dy, ha in [("NYJ", style.SERIES[1], "Jets", 7, 2, "left"),
                                ("LA", style.SERIES[0], "Rams", -7, 6, "right")]:
    ax.scatter(x[tm], y[tm], s=60, color=col, zorder=3, edgecolor=style.SURFACE, lw=0.8)
    ax.annotate(txt, (x[tm], y[tm]), xytext=(dx, dy), textcoords="offset points", ha=ha,
                fontsize=8.5, fontweight="bold", color=style.INK)
ax.set_xlabel("Average win probability on the team's snaps (%)")
ax.set_ylabel("Dropbacks, all snaps (%)")
ax.set_title("2025: the more a team trailed, the more it passed")
plt.show()
```

²Participation for 2023 on is supplied by FTN after the season ends, under a CC-BY-SA 4.0 licence. In season, nflverse's automated jobs rebuild play-by-play after each game window (Thursday, Sunday early and late, Sunday and Monday night) and again every morning, and check FTN for new charting every six hours; participation is rebuilt by hand once FTN delivers it. Workflow schedules: https://github.com/nflverse/nflverse-pbp/blob/master/.github/workflows/update_data.yaml, <https://github.com/nflverse/nflverse-fts/tree/main/.github/workflows>. Data dictionaries: https://nflreadr.com/articles/dictionary_participation.html, https://nflreadr.com/articles/dictionary_ftn_charting.html; release assets at <https://github.com/nflverse/nflverse-data/releases>.

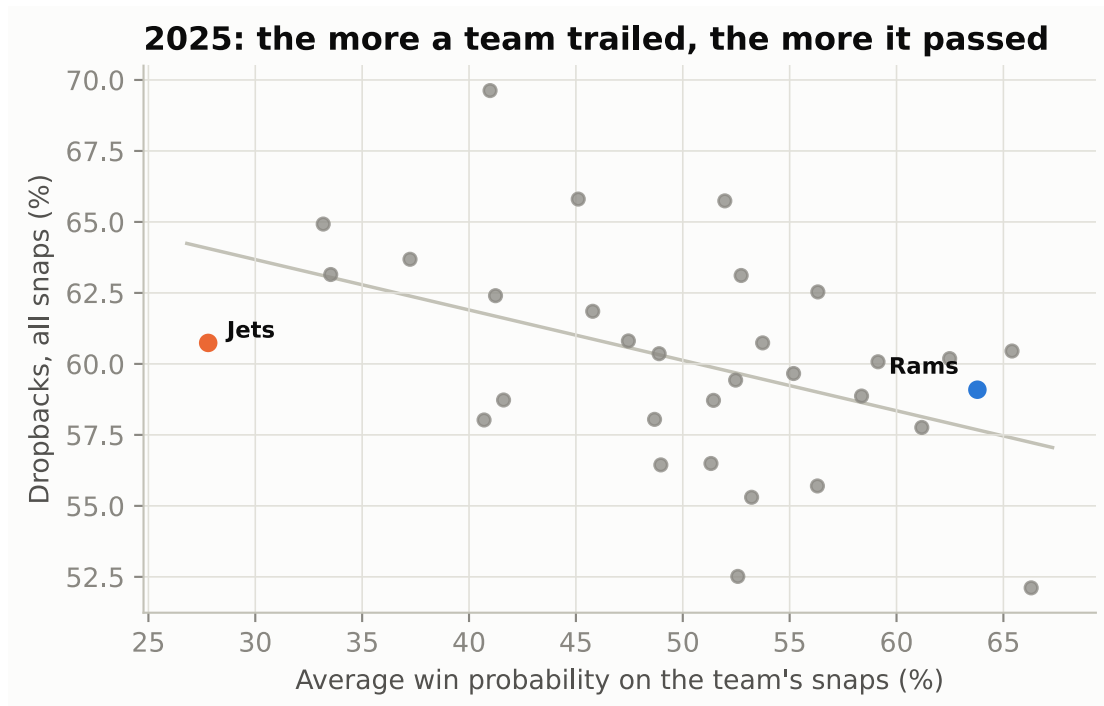


Figure 2: Every 2025 offense: how often it dropped back across all its regular-season snaps (up) against the average win probability on those snaps (across). The cloud slopes down: teams that were usually losing passed more, because trailing teams have to. The Jets (orange) and Rams (blue) passed at almost the same rate from opposite ends of the chart. Grey line: least-squares fit. Source: nflverse play-by-play.

The correlation is -0.45 . That is not a story about play-callers' tastes. It is clock arithmetic. A team down 14 in the fourth quarter must pass, because incompletions stop the clock and passes gain yards faster; a team up 14 runs, because every run keeps the clock moving and makes the other team's comeback harder. Add the downs (third-and-8 is a passing down for everybody) and the two-minute drills, and most of a raw pass rate is decided before the play-caller opens his call sheet. So when a broadcast says a team "is one of the most pass-happy offenses in the league," the first question is: or was it behind a lot?

🔥 Common misconception

"They threw it 650 times, so they're a passing team." Volume and rate both mostly follow the scoreboard. Bad teams throw a lot because they trail; good teams run a lot late because they lead. Before you call an offense pass-first or run-first, take the scoreboard and the clock out. The rest of this chapter shows two ways to do it: a filter and a model.

Neutral situations: taking the scoreboard out

The filter first. A **neutral situation** is a snap on which neither the scoreboard nor the clock forces the call. This chapter uses the definition that the play-calling chapters (Early Downs and Field Zones, 08-02 and 08-03) use: **win probability between 20% and 80%, and more than two minutes left in the half**. For play-calling tendencies, it usually adds **first and second down only**. Each piece has a reason:

- **Win probability 20–80%.** Inside that band the game is still in doubt, so a coach is not forced to throw (trailing badly) or to run out the clock (leading comfortably). Win probability is better than a score margin, because it knows that down 10 in the first quarter is a close game and down 10 with four minutes left is not.
- **Outside the last two minutes of each half.** The two-minute drill is its own sport: hurry-up passing, spiked balls, sideline routes. Every team plays it the same way.
- **First and second down.** Third down is ruled by the distance: almost everyone throws on third-and-8 and almost everyone runs on third-and-1. Early downs are where play-callers differ.

The first two filters are the garbage-time filter of 13-01 and 13-02 with a tighter band (they keep win probability 10–90%). How much do the exact cut-offs matter? Public analysts use several, so test it: rank every 2025 offense by its early-down pass rate under four definitions and see how far the rankings move.

```
s25 = plays[plays.season == 2025]
DEFS = {
    "Course: WP 20-80%, >2:00 left in half, downs 1-2": s25.early_neutral,
    "WP 10-90% instead": s25.wp.between(0.1, 0.9) & (s25.half_seconds_remaining > 120) & (s25.down <= 2),
    "Score within 7, first three quarters, downs 1-2": (s25.score_differential.abs() <= 7) & (s25.qtr <= 3)
    & (s25.down <= 2),
    "WP 20-80% and >2:00 left, all downs": s25.neutral,
}
rates = pd.DataFrame({k: s25[m].groupby("posteam")["pass"].mean() for k, m in DEFS.items()})
ranks = rates.rank()
course = list(DEFS)[0]
display(md_table(pd.DataFrame({
    "Neutral definition": list(DEFS),
    "Snaps kept": [m.mean() * 100 for m in DEFS.values()],
    "Rank correlation with the course's": [ranks[course].corr(ranks[k]) for k in DEFS],
})).round({"Snaps kept": 0})))
NEUTRAL_SHARE = s25.neutral.mean()
EARLY_SHARE = s25.early_neutral.mean()
RANK_R_SCORE = ranks[course].corr(ranks[list(DEFS)[2]])
assert min(ranks[course].corr(ranks[k]) for k in DEFS) >= 0.9
```

Neutral definition	Snaps kept	Rank correlation with the course's
Course: WP 20-80%, >2:00 left in half, downs 1-2	43	1.00
WP 10-90% instead	53	0.96
Score within 7, first three quarters, downs 1-2	40	0.92
WP 20-80% and >2:00 left, all downs	56	0.97

The course's filter keeps 43% of all 2025 scrimmage snaps (and 56% if third and fourth downs stay in). The rankings barely care which reasonable version you use: the rank correlations are 0.9 or higher (a "score within 7" filter gives 0.92). That's reassuring, and it is also a warning. The filter is crude. It still mixes second-and-2 with second-and-15, the opponent's 5-yard line with your own 20, and a 7-point lead with a 7-point deficit. A team whose early downs happened to be second-and-long more often will look pass-happy in a neutral pass rate for reasons that have nothing to do with its coach. Fixing that needs a model.

Expected pass rate: the league-average play-caller

Here is the idea. For every snap, ask: **in exactly this situation, how often would an average NFL offense drop back to pass?** That probability is the snap's **expected pass rate**, or **xpass**. It is to play-calling what expected points are to scoring in 13-02: a baseline that prices the situation, so that what is left over belongs to the team. Third-and-12 is close to a certain dropback; second-and-1 at midfield in a tied game leans hard toward a run; first-and-10 down 17 in the fourth quarter leans hard toward a pass.

nflverse ships an xpass for every snap since 2006 (the first season in which the play-by-play marks scrambles), from a gradient-boosted tree model (the XGBoost library) that Ben Baldwin added to the open-source nflfastR project in 2020.³ You can use it directly, and the rest of this chapter mostly does. But it is worth fitting a small one yourself, for the same reason 13-02 fitted its own expected-points model: once you have built it, you know what it can and cannot see. A **logistic regression** is enough: a model that turns a weighted sum of situation features into a probability between 0 and 1, fitted by pushing down the **log loss** (the average of $-\log$ of the probability the model gave to what actually happened).

The features, and why each is there:

³nflfastR reference, "Compute expected dropback probability" (`add_xpass()`, which adds `xpass` and `pass_oe`): https://www.nflfastR.com/reference/add_xpass.html; play-by-play field definitions at https://nflreadr.nflverse.com/articles/dictionary_pbp.html. The model's source (`R/helper_add_xpass.R`, author Ben Baldwin) returns NA before 2006, "since that was before NFL started marking scrambles", and feeds the model down, yards to go, yard line, quarter, win probability, Vegas-adjusted win probability, era, score differential, home, seconds left in the half, both teams' timeouts and roof type (outdoors, retractable, dome): https://github.com/nflverse/nflfastR/blob/master/R/helper_add_xpass.R. The model is stored as an XGBoost model in the `fastrmodels` package (<https://github.com/nflverse/fastrmodels>); `add_xpass()` first shipped in nflfastR 3.1.1 (October 2020), described in its NEWS as "our experimental expected pass model".

- **Down, and distance by down.** Distance enters on a log scale, separately for each down, because 1 versus 3 yards to go matters more than 11 versus 13, and because third-and-6 is a passing down while first-and-6 is not.
- **Field position, with knots.** Like 13-02's EP model, the yard line gets extra "how far past this point" terms near both goal lines (the 10, the 20, your own 20 and 10), so the curve can bend where play-calling changes: tight red-zone windows, and backed-up offenses that run to get breathing room.
- **Score, scaled by how much of the game has gone.** A 10-point deficit is a mild nudge toward passing in the first quarter and an order in the fourth. So the score enters on its own and multiplied by the share of the game played, once in a straight line and once raised to the fourth power, so that it bites hardest at the end. Deficits get their own terms, because trailing and leading are not mirror images (leading teams run, but trailing teams pass harder).
- **The end of each half.** Inside two minutes of the first half everyone throws; inside two minutes of the second, the trailing team throws and the leading team kneels or runs.

```
def xp_features(s, situation_only=False):
    """Situation -> features for an expected-pass model."""
    dn = s["down"].to_numpy()
    yl = s["yardline_100"].to_numpy() / 100
    ytg = np.log(np.clip(s["ydstogo"].to_numpy(), 1, None))
    cols = [np.ones_like(yl)]
    cols += [(dn == k).astype(float) for k in (2, 3, 4)] # down
    cols += [ytg * (dn == k) for k in (1, 2, 3, 4)] # distance, by down
    cols += [yl] + [np.clip(yl - k, 0, None) for k in (0.1, 0.2, 0.8, 0.9)] # field position
    if not situation_only:
        sd = np.clip(s["score_differential"].to_numpy(), -24, 24) / 10 # score, in tens
        gone = 1 - s["game_seconds_remaining"].to_numpy() / 3600 # share of game played
        late = (s["half_seconds_remaining"].to_numpy() <= 120).astype(float)
        first_half = (s["qtr"].to_numpy() <= 2).astype(float)
        behind = np.minimum(sd, 0)
        cols += [sd, sd * gone, behind * gone, sd * gone**4, behind * gone**4, np.abs(sd) * gone**4]
        cols += [late * first_half, late * (1 - first_half) * (sd < 0), late * (1 - first_half) * (sd > 0)]
    return np.column_stack(cols).astype(np.float32)

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

def fit_logistic(X, y, steps=600, lr=0.05):
    """Logistic regression by gradient descent on the log loss (Adam step sizes, as in 13-02).
    Returns a function that maps a feature matrix to probabilities."""
    mu, sd = X.mean(0), X.std(0)
    mu[0], sd[0] = 0, 1 # leave the intercept alone
    Xs = (X - mu) / sd
    w = np.zeros(X.shape[1], dtype=np.float32)
    m, v = np.zeros_like(w), np.zeros_like(w)
    for i in range(1, steps + 1):
        g = Xs.T @ (sigmoid(Xs @ w) - y) / len(y)
        m, v = 0.9 * m + 0.1 * g, 0.999 * v + 0.001 * g**2
        w -= lr * (m / (1 - 0.9**i)) / (np.sqrt(v / (1 - 0.999**i)) + 1e-8)
    return lambda Z: sigmoid(((Z - mu) / sd) @ w)

train, test = plays[plays.season <= 2024], plays[plays.season == 2025]
y_train = train["pass"].to_numpy(np.float32)
y_test = test["pass"].to_numpy()
situation_only = fit_logistic(xp_features(train, True), y_train)
```

```

xpass_model = fit_logistic(xp_features(train), y_train)

def log_loss(p, y):
    p = np.clip(p, 1e-6, 1 - 1e-6)
    return -np.mean(y * np.log(p) + (1 - y) * np.log(1 - p))

def auc(p, y):
    """Chance that a random dropback gets a higher xpass than a random designed run."""
    rank = pd.Series(p).rank().to_numpy() # tied predictions share a rank
    n1 = y.sum()
    return (rank[y == 1].sum() - n1 * (n1 + 1) / 2) / (n1 * (len(y) - n1))

has_nflverse = test.xpass.notna().to_numpy()
CANDIDATES = {
    "League pass rate for every snap": np.full(len(test), y_train.mean()),
    "Down, distance and field position only": situation_only(xp_features(test, True)),
    "Plus score and clock (this chapter's model)": xpass_model(xp_features(test)),
    "nflverse xpass (XGBoost)": test.xpass.to_numpy(),
}
display(md_table(pd.DataFrame(
    [(k, log_loss(p[has_nflverse], y_test[has_nflverse]), auc(p[has_nflverse], y_test[has_nflverse]))
    for k, p in CANDIDATES.items()], columns=["Model (fitted 2016-2024, scored on 2025)", "Log loss", "AUC"],
    digits=3))
plays["xpass_ours"] = xpass_model(xp_features(plays))
plays["proe_ours"] = 100 * (plays["pass"] - plays.xpass_ours)

```

Model (fitted 2016-2024, scored on 2025)	Log loss	AUC
League pass rate for every snap	0.673	0.500
Down, distance and field position only	0.595	0.720
Plus score and clock (this chapter's model)	0.562	0.769
nflverse xpass (XGBoost)	0.536	0.786

Read the table from the top. Guessing the league's pass rate for every snap is the floor. Down, distance and field position alone recover much of the signal. Adding score and clock gets most of the rest: the model ranks a random 2025 dropback above a random 2025 run about three times in four (the **AUC**, short for "area under the curve", is exactly that chance; 0.5 is a coin flip, 1.0 perfect). nflverse's model is better still, because it has more features and a more flexible shape, but ours is close, and it was scored on a season it never saw.

```

def situation(down, ydstogo, yardline_100, score_differential, game_seconds_remaining, qtr):
    n = np.broadcast(down, ydstogo, yardline_100, score_differential, game_seconds_remaining).shape or (1,)
    half = np.where(np.asarray(qtr) <= 2, np.asarray(game_seconds_remaining) - 1800,
                    np.asarray(game_seconds_remaining))
    return pd.DataFrame({k: np.broadcast_to(np.asarray(v, dtype=float), n) for k, v in dict(
        down=down, ydstogo=ydstogo, yardline_100=yardline_100, score_differential=score_differential,
        game_seconds_remaining=game_seconds_remaining, half_seconds_remaining=half, qtr=qtr).items()})

fig, (a1, a2) = plt.subplots(1, 2, figsize=(6.6, 3.3), sharey=True)
score = np.arange(-21, 22)

```

```

MOMENTS = [("Opening kickoff", 3600, 1), ("Start of 3rd quarter", 1800, 3),
           ("8:00 left", 480, 4), ("3:00 left", 180, 4)]
for k, (lab, secs, q) in enumerate(MOMENTS):
    a1.plot(score, 100 * xpass_model(xp_features(situation(1, 10, 75, score, secs, q))),
           color=style.SEQUENTIAL_BLUE[[2, 3, 5, 6][k]], lw=2, label=lab)
a1.axvline(0, color=style.BASELINE, lw=1)
a1.set_xlabel("Offense's score margin")
a1.set_ylabel("Expected pass rate (%)")
a1.set_title("1st & 10, own 25", fontsize=9.5)
a1.set_ylim(0, 100)
a1.legend(fontsize=7, loc="upper right", handlelength=1.2)
togo = np.arange(1, 16)
for k, dn in enumerate([1, 2, 3]):
    p = 100 * xpass_model(xp_features(situation(dn, togo, 50, 0, 2400, 2)))
    a2.plot(togo, p, color=style.SERIES[k], lw=2)
    a2.text(togo[-1] + 0.3, p[-1], ["1st", "2nd", "3rd"][k] + " down", fontsize=7.5,
           color=style.INK, va="center")
a2.set_xlim(1, 19.5)
a2.set_xticks([1, 5, 10, 15])
a2.set_xlabel("Yards to go")
a2.set_title("Tied, 2nd quarter, midfield", fontsize=9.5)
fig.tight_layout(w_pad=1.0)
plt.show()
XP_DOWN17 = xpass_model(xp_features(situation(1, 10, 75, -17, 300, 4)))[0]

```

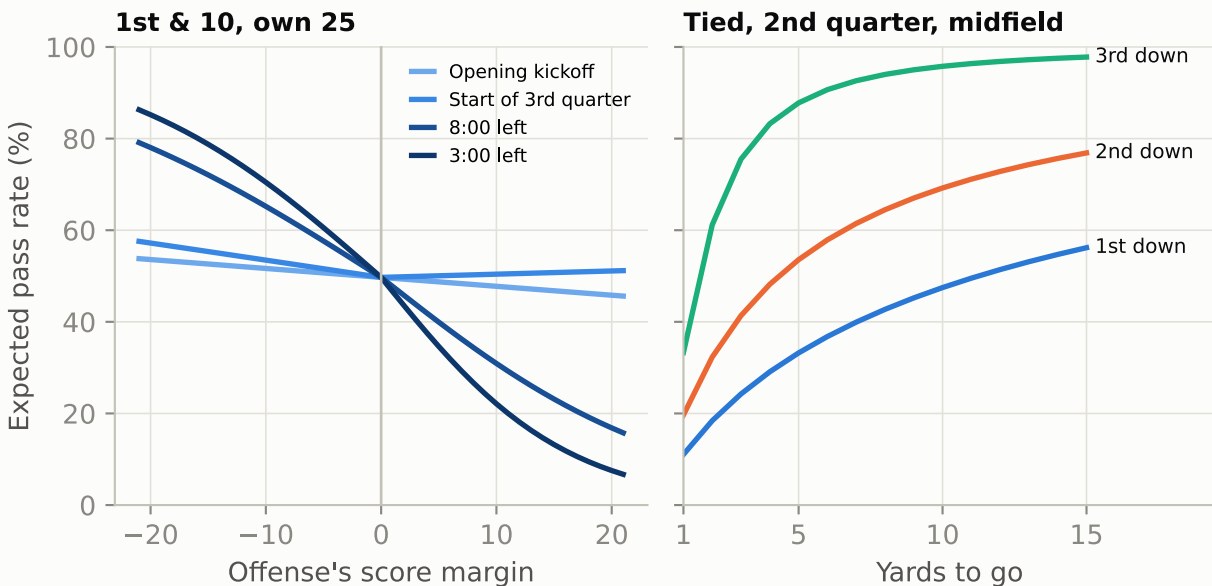


Figure 3: What the expected-pass model has learned (this chapter's model). Left: first-and-10 at the offense's own 25, by the score, at four moments of the game. Early on the score barely matters; with three minutes left, a team down three scores is expected to throw on more than four snaps in five and a team up three scores on fewer than one in ten. Right: tied game, second quarter, ball at midfield, by down and yards to go. First-and-10 is close to a coin flip, and the model treats short first downs (rare outside goal-to-go and penalties) as run downs; third down becomes a passing down within a few yards.

! Why it exists

A team's pass rate is the sum of two things: the situations it faced and what it chose to do in them. Coaches have always known this, which is why tendency reports are filed by down, distance and field zone (Game Planning Week). xpass does the same job in one number. It prices every snap's situation, so a team's average xpass is the pass rate an average play-caller would have produced from *its* schedule of situations. The trade: xpass knows only what you give it. Ours ignores the betting line, the stadium, injuries and the opponent; nflverse's adds the betting line (through a Vegas-adjusted win probability), timeouts, home field and roof type, but still sees no personnel, no quarterback and no opposing defense. That is deliberate, since those are the things you want to show up in the leftover, but it means the leftover holds roster and opponent as well as the play-caller's taste.

PROE: the choice, measured

Now subtract. For each snap, take what happened (1 for a dropback, 0 for a run) minus its xpass. Average that over a team's snaps and multiply by 100, and you have **pass rate over expected**, or **PROE**, in percentage points:

$$\text{PROE} = \frac{100}{n} \sum_{i=1}^n (\text{pass}_i - \text{xpass}_i) = 100 (\text{actual pass rate} - \text{average xpass})$$

A PROE of +5 means the team dropped back five points more often than an average play-caller would have in the same situations; -10 means it ran ten points more often. nflverse stores the per-snap version as `pass_oe`, already multiplied by 100.

Why still use the neutral filter, when xpass already prices the situation? Because PROE is meant to measure *choice*, and outside neutral snaps there is little choice to measure. Down 17 with five minutes left, xpass is about 0.79 and nearly every team passes; averaging in hundreds of snaps like that adds noise and almost no information. The model is also least trustworthy in those corners, where it has the fewest examples. So the convention, used here and, with small variations in the cut-offs, by most public analysts, is **PROE on neutral first and second downs**.

```
EN = plays[plays.early_neutral]
TEAM = EN.groupby(["season", "posteam"]).agg(
    snaps=("pass", "size"), pass_rate=("pass", "mean"), xpass=("xpass", "mean"),
    proe=("pass_oe", "mean"), proe_ours=("proe_ours", "mean"))
TEAM["epa"] = plays[plays.neutral].groupby(["season", "posteam"]).epa.mean() # EPA/play, neutral
TEAM = TEAM.join(RAW)
T25 = TEAM.loc[2025].copy()
T25["raw_rank"] = T25.raw.rank(ascending=False).astype(int)
T25["proe_rank"] = T25.proe.rank(ascending=False).astype(int)
R_OURS = TEAM.proe.corr(TEAM.proe_ours)
SD_OE = EN.pass_oe.std() # how much one snap's "pass minus xpass" swings

show = pd.concat([T25.nlargest(5, "proe"), T25.nsmallest(5, "proe").iloc[::-1]])
display(md_table(pd.DataFrame({
    "Team": show.index,
    "Pass %, all snaps": show.raw * 100, "Rank": show.raw_rank,
    "Pass %, neutral": show.pass_rate * 100, "xpass %": show.xpass * 100,
    "PROE": show.proe, "PROE, ours": show.proe_ours}), digits=1))
```

Team	Pass %, all snaps	Rank	Pass %, neutral	xpass %	PROE	PROE, ours
ARI	69.6	1	60.9	52.9	8.0	7.9
LA	59.1	20	58.0	52.2	5.9	7.2
CIN	65.8	2	58.4	53.5	4.9	5.1
KC	65.7	3	58.4	53.6	4.7	6.2

Team	Pass %, all snaps	Rank	Pass %, neutral	xpass %	PROE	PROE, ours
JAX	60.2	16	55.6	51.4	4.2	4.3
DET	59.4	19	49.2	54.6	-5.4	-3.3
GB	55.7	29	48.2	53.9	-5.7	-4.3
ATL	56.5	27	48.3	54.2	-5.9	-4.9
BAL	52.5	31	42.9	54.1	-11.2	-9.9
NYJ	60.7	13	40.8	53.0	-12.2	-13.2

The table shows the five most pass-leaning and five most run-leaning offenses of 2025 on neutral early downs, from the top. The two PROE columns agree closely: across all 320 team-seasons from 2016 to 2025, our model's PROE and nflverse's correlate at 0.98, so the small model is good enough for this job. The interesting column is the raw rank beside them. The Rams, second in PROE, were 20th in raw pass rate; the Jets, last in PROE, were 13th.

Film room: New York Jets and Los Angeles Rams 2025

Put the opening's two offenses through the whole pipeline. Across all their snaps, the Jets passed on 61% and the Rams on 59%. Filter to neutral early downs and the Jets fall to 41% while the Rams barely move (58%). Price the situations and the gap widens: the Jets' PROE was -12.2, the lowest in the league, and the Rams' +5.9, second only to Arizona.

What to notice: the two raw rates were almost equal for opposite reasons. The Jets spent the season behind (average win probability on their snaps 28%), so the clock made them throw; when they had a choice, they ran. The Rams spent it ahead (64%), so the clock made them run late; when they had a choice, they put the ball in their MVP quarterback's hands. Neither raw number told you anything about either coaching staff. PROE did.

PROE by situation

A season PROE is an average over very different moments. Splitting it is where a scouting report starts: on which downs, scores and parts of the field does this team break from the league? The recipe is the same subtraction on a subset of snaps, and Figure 4 applies it to two teams near opposite ends of the 2025 table: the Rams, second from the top, and the Ravens, second from the bottom.

```
SITS = {
    "1st & 10": lambda d: (d.down == 1) & (d.ydstogo == 10),
    "2nd & 1-6": lambda d: (d.down == 2) & (d.ydstogo <= 6),
    "2nd & 7+": lambda d: (d.down == 2) & (d.ydstogo >= 7),
    "3rd & 1-3": lambda d: (d.down == 3) & (d.ydstogo <= 3),
    "3rd & 4+": lambda d: (d.down == 3) & (d.ydstogo >= 4),
    "Leading": lambda d: d.score_differential > 0,
    "Tied": lambda d: d.score_differential == 0,
    "Trailing": lambda d: d.score_differential < 0,
    "Red zone": lambda d: d.yardline_100 <= 20,
}

def proe_by_situation(team, season):
    d = plays[plays.neutral & (plays.season == season) & (plays.posteam == team) & plays.pass_oe.notna()]
    rows = []
    for name, rule in SITS.items():
        x = d[rule(d)].pass_oe
        rows.append((name, x.mean(), x.std() / np.sqrt(len(x)), len(x)))
    return pd.DataFrame(rows, columns=["situation", "proe", "se", "n"]).set_index("situation")
```

```

PS = {tm: proe_by_situation(tm, 2025) for tm in ["LA", "BAL"]}
fig, ax = plt.subplots(figsize=(6.4, 3.9))
yy = np.arange(len(SITS))[:-1]
for k, (tm, lab) in enumerate([("LA", "Rams"), ("BAL", "Ravens")]):
    t = PS[tm]
    off_y = yy + (0.17 if k == 0 else -0.17)
    col = style.SERIES[0] if k == 0 else style.SERIES[1]
    ax.hlines(off_y, t.proe - t.se, t.proe + t.se, color=col, lw=1.4, alpha=0.6)
    ax.scatter(t.proe, off_y, s=34, color=col, zorder=3, label=lab, edgecolor=style.SURFACE, lw=0.6)
    for yv, n in zip(off_y, t.n):
        ax.text(33 + 8 * k, yv, f"{n}", fontsize=6.6, color=col, va="center", ha="right")
ax.text(37, yy[0] + 0.62, "snaps (n)", fontsize=6.8, color=style.INK_2, ha="center")
ax.axvline(0, color=style.INK_2, lw=1)
ax.set_yticks(yy, list(SITS), fontsize=8)
ax.set_xlim(-42, 43)
ax.set_ylim(-0.6, len(SITS) - 0.1)
ax.set_xticks([-40, -30, -20, -10, 0, 10, 20])
ax.set_xlabel("Pass rate over expected (percentage points)")
ax.grid(axis="y", visible=False)
ax.legend(loc="lower left", fontsize=8)
ax.set_title("Where the Rams and Ravens left the script, 2025")
plt.show()

```

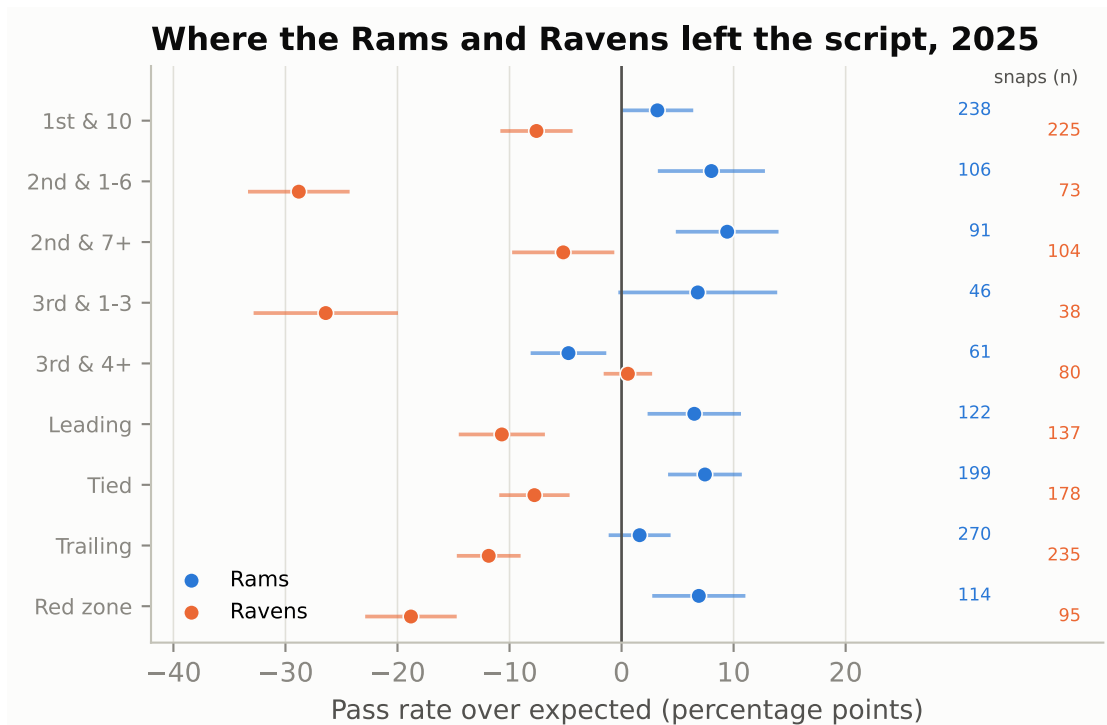


Figure 4: PROE by situation, 2025 regular season, neutral snaps (win probability 20-80%, outside the last two minutes of each half; all downs, so third down appears). Each dot is the team's pass rate over expected in that situation, in percentage points; the whisker is plus or minus one standard error, the noise you'd expect from that many snaps (n, right). The Rams passed more than expected in nearly every situation; the Ravens ran more than expected in nearly every one, most of all in short yardage and the red zone. On third-and-4 or more xpass is already high, so there is less room to deviate.

The whiskers are the lesson as much as the dots. A team's whole-season PROE rests on four or five hundred neutral snaps, but a situation like "third-and-1 to 3" holds forty or so, and one standard error there is about seven points. The general rule: the standard

error of a PROE is roughly 48 divided by the square root of the number of snaps, because each snap's "pass minus xpass" swings by about 48 points (its standard deviation on neutral early downs). A hundred snaps gives about ± 4.8 points; twenty-five, about ± 10 . Read small cells as hints, not findings.

Does passing more pay? PROE versus EPA

13-02 and Early Downs and Field Zones showed that on early downs, league-wide, a dropback earns more EPA than a run. That is the analytics argument for passing more, and PROE is how the argument gets aimed at a single team. Does it show up team by team?

```
R_ALL = TEAM.proe.corr(TEAM.epa)
PROE_RANK = TEAM.groupby(level="season").proe.rank(ascending=False)
assert all(PROE_RANK.loc[(s, "CIN")] <= 3 for s in (2022, 2023, 2024))
R_BY_SEASON = TEAM.groupby(level="season").apply(lambda t: t.proe.corr(t.epa))

fig, ax = plt.subplots(figsize=(6.4, 4.2))
old = TEAM.loc[TEAM.index.get_level_values("season") <= 2024]
ax.scatter(old.proe, old.epa, s=10, color=style.GRID, edgecolor=style.BASLINE, lw=0.3, zorder=1)
ax.scatter(T25.proe, T25.epa, s=26, color=style.SERIES[0], zorder=3, edgecolor=style.SURFACE, lw=0.6)
LABELS = {"LA": (8, 3, "left"), "NE": (8, 0, "left"), "ARI": (8, 0, "left"), "NYJ": (8, 0, "left"),
          "BAL": (8, -5, "left"), "CIN": (8, -7, "left"), "KC": (-8, -9, "right"), "BUF": (-8, 6, "right"),
          "DET": (-14, 12, "right"), "CAR": (8, -4, "left")}
for tm, (dx, dy, ha) in LABELS.items():
    ax.annotate(tm, (T25.loc[tm, "proe"], T25.loc[tm, "epa"]), xytext=(dx, dy), textcoords="offset points",
               ha=ha, va="center", fontsize=7.5, color=style.INK,
               arrowprops=dict(arrowstyle="-", color=style.INK_2, lw=0.5, shrinkA=1, shrinkB=3))
ax.axhline(0, color=style.BASLINE, lw=1)
ax.axvline(0, color=style.BASLINE, lw=1)
ax.set_xlabel("PROE, neutral 1st & 2nd down (percentage points)")
ax.set_ylabel("EPA per play, neutral snaps")
ax.set_title("Passing more than expected, and scoring")
plt.show()
```

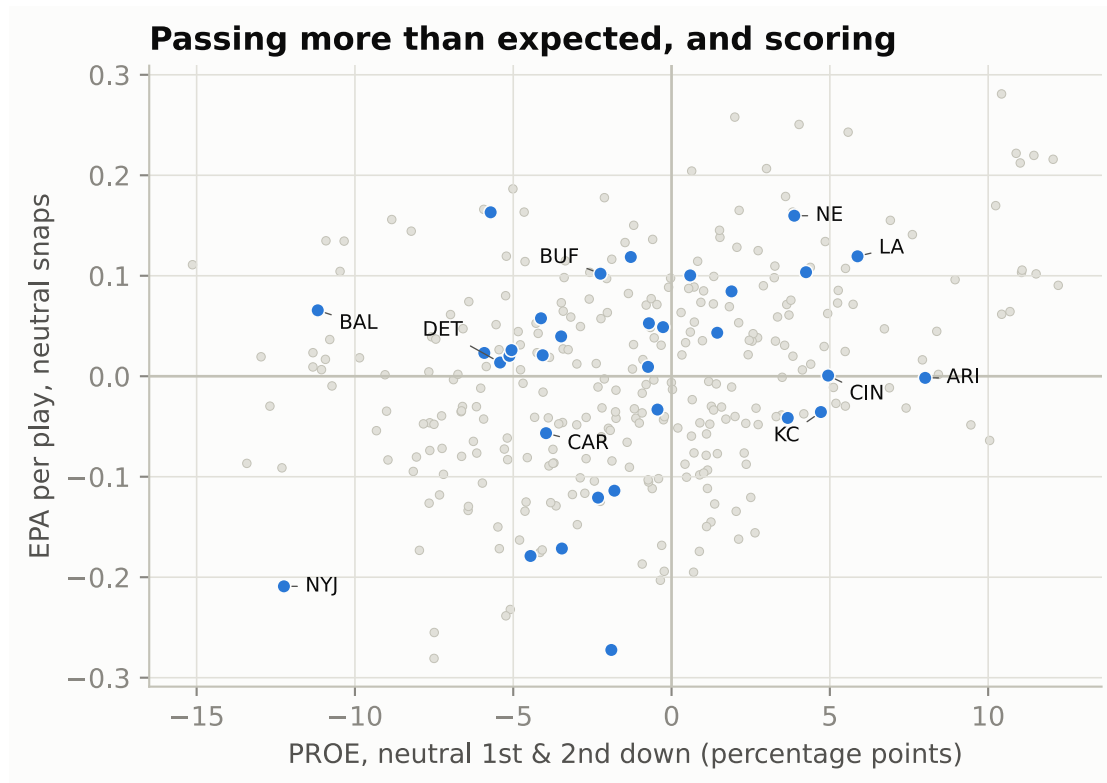


Figure 5: Team PROE against offensive efficiency. Each grey dot is one offense in one season, 2016-2024; blue dots are 2025, with selected teams labelled. Across: PROE on neutral early downs. Up: EPA per play on all neutral snaps. Passing more than expected goes with scoring more, but loosely: the correlation across all 320 team-seasons is about 0.3, and some of the best offenses sit left of zero. Source: nflverse play-by-play (xpass and pass_oe from the nflfastR model).

The slope is real but gentle. Across 2016–2025 the correlation is 0.31; in single seasons it has ranged from 0.05 to 0.51 (0.05 in 2024, 0.27 in 2025). And even that gentle slope can’t be read as “pass more and you’ll score more,” because the arrow runs both ways:

- **Good passers earn the right to pass.** A team with an elite quarterback throws more *because* throwing works for it. The 2022–2024 Bengals were among the league’s three most pass-leaning offenses every season, built around Joe Burrow; that PROE is partly a description of their roster.
- **Struggling passers get hidden.** A coach with a poor or injured quarterback, or a bad pass protection, runs more on purpose. Low PROE and low EPA then share a cause.
- **The league-wide edge is an average.** Dropbacks beat runs on early downs on average, but a team’s runs can be better than average (an elite line and back) and its passes worse.

So treat PROE as what it is: a description of *style*, measured cleanly. It tells a defense what to expect. It doesn’t grade the coach.

⁴Super Bowl LIX, February 9, 2025, Caesars Superdome: Philadelphia 40, Kansas City 22 (course fact sheet; <https://www.cbsnews.com/amp/news/nfl-mvp-super-bowl-jalen-hurts-eagles>). Barkley’s 2,005 regular-season rushing yards: Pro Football Reference, <https://www.pro-football-reference.com/players/B/BarkSa00.htm>. Kellen Moore hired as offensive coordinator and play-caller in February 2024, replacing Brian Johnson: <https://www.inquirer.com/eagles/eagles-hire-kellen-moore-offensive-coordinator-20240205.html>; Kevin Patullo was the 2025 coordinator (course fact sheet).

i Film room: Philadelphia Eagles 2024

The 2024 Eagles are the cleanest counter-example to “pass more.” On neutral early downs their PROE was -8.2 , 2nd-lowest in the league, after two seasons near $+5$ ($+4.9$ in 2022, $+4.4$ in 2023). The change had two names: Saquon Barkley, who ran for 2,005 yards behind one of the league’s best offensive lines, and a new play-caller, offensive coordinator Kellen Moore. Philadelphia’s offense was 4th in neutral EPA per play, and the season ended with a 40–22 win over Kansas City in Super Bowl LIX.⁴

What to notice: PROE moved by about thirteen points in one year with the same head coach and quarterback, because a new back and a new play-caller changed what was worth doing. And the Eagles’ 2025 PROE, under yet another coordinator (Kevin Patullo), came back toward zero (-0.3). A team’s PROE is an answer to its own personnel and play-caller, not a fixed belief.

🔥 Common misconception

“Negative PROE means an old-fashioned, analytically backward coach.” Sometimes. But PROE also falls when a team has a dominant running game, a hurt quarterback, a lead it is protecting within the neutral band, or a quarterback whose scrambles turn called passes into runs on the stat sheet (they don’t: nflverse counts scrambles as dropbacks, which is exactly why). Read PROE as “what to expect,” then look for the reason before you judge it.

How long does a tendency last?

A tendency is only useful to an opponent if it persists: from the games already played to the next one, and from last season into this one. PROE turns out to be one of the steadiest numbers in football, steadier than efficiency.

```
n_ = EN.copy()
n_["game_no"] = n_.groupby(["season", "posteam"]).week.rank(method="dense")
CARRY = []
for k in range(1, 9):
    first = n_[n_.game_no <= k].groupby(["season", "posteam"])[["pass_oe", "epa"]].mean()
    rest = n_[n_.game_no > k].groupby(["season", "posteam"])[["pass_oe", "epa"]].mean()
    CARRY.append((k, first.pass_oe.corr(rest.pass_oe), first.epa.corr(rest.epa)))
CARRY = pd.DataFrame(CARRY, columns=["k", "proe", "epa"]).set_index("k")

# Head coach of record: the coach in charge for most of each team's regular-season games.
OLD_ABBR = {"OAK": "LV", "SD": "LAC", "STL": "LA"}
sch = nfl().load_schedules(list(SEASONS)).to_pandas()
sch = sch[sch.game_type == "REG"]
hc = pd.concat([sch[["season", t, f"{side}_coach"]].set_axis(["season", "team", "coach"], axis=1)
               for side, t in (("home", "home_team"), ("away", "away_team"))])
hc["team"] = hc.team.replace(OLD_ABBR)
HC = hc.groupby(["season", "team"]).coach.agg(lambda s: s.value_counts().index[0])

yoy = TEAM[["proe", "epa"]].reset_index().rename(columns={"posteam": "team"})
yoy["team"] = yoy.team.replace(OLD_ABBR)
yoy = yoy.merge(yoy.assign(season=yoy.season + 1), on=["season", "team"], suffixes=("", "_prev"))
yoy["same_hc"] = [HC.get((s, t)) == HC.get((s - 1, t)) for s, t in zip(yoy.season, yoy.team)]
R_YOY = {same: (g.proe.corr(g.proe_prev), g.epa.corr(g.epa_prev), len(g)) for same, g in
yoy.groupby("same_hc")}
assert PROE_RANK.loc[(2023, "ARI")] >= 27 and PROE_RANK.loc[(2025, "ARI")] <= 3 # quoted in the text

fig, (a1, a2) = plt.subplots(1, 2, figsize=(6.6, 3.2), gridspec_kw={"width_ratios": [1, 1.1]})
a1.plot(CARRY.index, CARRY.proe, color=style.SERIES[0], marker="o", ms=3.5, lw=2)
a1.plot(CARRY.index, CARRY.epa, color=style.MUTED, marker="o", ms=3.5, lw=2)
```

```

a1.text(8.15, CARRY.proe.iloc[-1], "PROE", fontsize=7.5, color=style.SERIES[0], va="center")
a1.text(8.15, CARRY.epa.iloc[-1], "EPA/play", fontsize=7.5, color=style.INK_2, va="center")
a1.set_xlim(0.7, 10.2)
a1.set_ylim(0, 0.65)
a1.set_xticks(range(1, 9))
a1.set_xlabel("Games played so far (k)")
a1.set_ylabel("Correlation with rest of season")
a1.set_title("Within a season", fontsize=9.5)
for same, col, lab in [(True, style.SERIES[0], "same head coach"), (False, style.SERIES[1], "new head
coach")]:
    g = yoy[yoy.same_hc == same]
    a2.scatter(g.proe_prev, g.proe, s=10 if same else 16, color=col, alpha=0.55 if same else 0.9,
               edgecolor="none", label=f"{lab} (r = {R_YOY[same][0]:.2f})")
a2.axhline(0, color=style.BASELINE, lw=0.8)
a2.axvline(0, color=style.BASELINE, lw=0.8)
a2.set_xlabel("PROE, previous season")
a2.set_ylabel("PROE, this season")
a2.set_title("Season to season", fontsize=9.5)
a2.legend(fontsize=7, loc="upper center", bbox_to_anchor=(0.45, -0.22), ncol=1, handletextpad=0.2)
fig.tight_layout(w_pad=1.2)
plt.show()

```

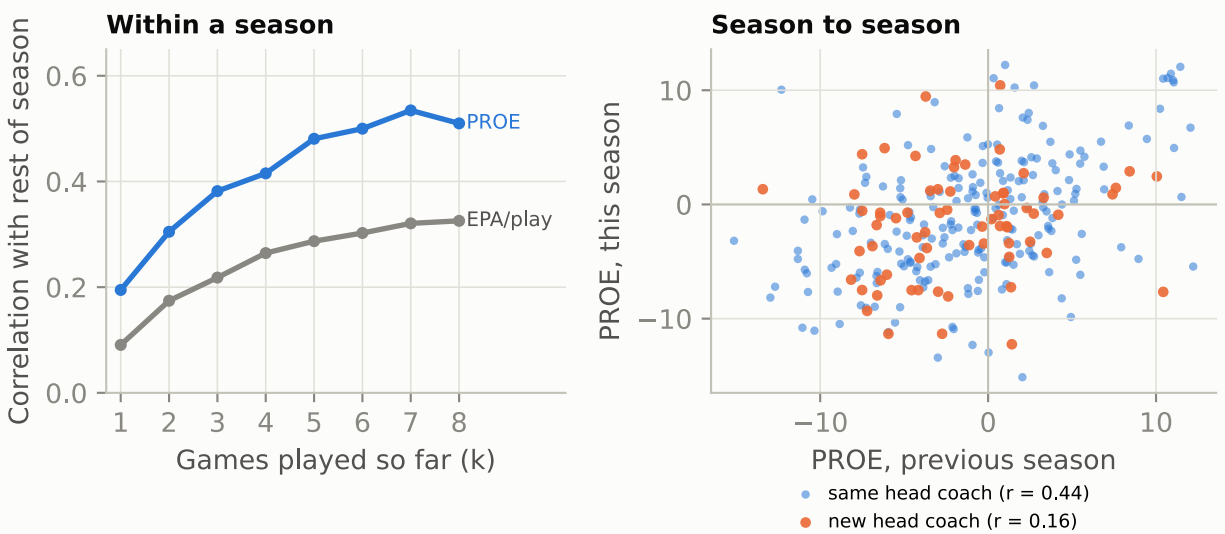


Figure 6: How well PROE carries forward, 2016-2025 regular seasons, neutral early downs. Left: correlation between a team's numbers over its first k games and over the rest of the season, for PROE (blue) and EPA per play (grey). After four games PROE already predicts the rest of the season better than EPA does after eight. Right: each team's PROE one season against the next, split by whether the main head coach stayed (blue) or changed (orange). With the same head coach, PROE carries over; with a new one, much less.

Two practical rules come out of that chart.

Within a season, trust PROE early. After four games, a team's PROE correlates 0.42 with its PROE over the rest of the season; EPA per play manages only 0.26, and even after eight games EPA (0.33) has not caught up with four games of PROE. The reason is the same one that makes 13-02's success rate steadier than EPA: a play *call* is a choice the coach makes every snap, while a play's *result* carries the whole randomness of football (a tipped ball, a missed tackle, a fumble). Style shows up faster than quality.

Across seasons, PROE belongs to the coach. When the head coach stayed, PROE carried over from one season to the next with a correlation of 0.44 (220 team-seasons). When he changed, it fell to 0.16 (68). Head coach is a rough stand-in for play-caller (some head coaches call the defense or hand the offense to a coordinator), and a cleaner test would track the play-caller himself. Game Planning Week did that for one man, Drew Petzing, whose Arizona offenses went from among the most run-leaning in 2023 to

among the most pass-leaning by 2025, under the same play-caller. A new play-caller resets the prior almost completely, and that matters right now. For 2026, Seattle (Brian Fleury), Detroit (Drew Petzing), Kansas City (Eric Bieniemy), Baltimore (Declan Doyle, under new head coach Jesse Minter) and Philadelphia (Sean Mannion) all have new offensive coordinators, among others.⁵ Their 2025 PROE is a weak guide to their 2026 offense; the first four 2026 games are a better one.

Personnel, alignment, motion and play-action

PROE answers “run or pass, compared with the situation?” A defense preparing for a game wants the next layer down: *from what?* Constraint Theory and Sequencing built a tendency table for one team, pass rate by personnel and quarterback alignment, and showed that the alignment alone calls about two snaps in three league-wide. Here the same ingredients become league-wide measures, one number per team, so you can see at a glance where an offense differs from the other 31. Each is measured on neutral early downs, and each has a reason a defensive coordinator cares:

- **Personnel shares** (11, 12, and “heavy”: two or more backs or three or more tight ends). Personnel decides the defense’s own personnel. Heavy groupings pull a defense into its base package or a bigger front, and a team that lives in them dictates matchups (Personnel Groupings).
- **Under-center share**, and the **pass rate from under center**. The quarterback’s alignment is the loudest run/pass tell in public data. A team that throws a lot from under center has neutralised it (see 08-02).
- **Motion rate**, from FTN’s `is_motion`. Motion before or at the snap forces the defense to adjust and communicate, and can reveal man or zone (Motion and Shifts).
- **Play-action rate**, as a share of dropbacks. Play-action depends on the defense believing the run, so it pairs with heavy personnel and under-center snaps (Play-Action, Boots and Screens).
- **No-huddle rate**. Lining up without a huddle limits the defense’s substitutions and its time to disguise (In-Game Adjustments and Tempo).

```
def offense_metrics(season):
    """One row per offense: neutral early-down tendencies from participation and FTN charting."""
    o = snaps[(snaps.season == season) & snaps.early_neutral]
    t = o.groupby("posteam").agg(
        snaps=("pass", "size"), proe=("pass_oe", "mean"),
        p11=("pers", lambda s: (s == "11").mean()), p12=("pers", lambda s: (s == "12").mean()),
        heavy=("heavy", "mean"), uc=("uc", "mean"), motion=("is_motion", "mean"),
        nohuddle=("is_no_huddle", "mean"))
    t["uc_pass"] = o[o.uc == 1].groupby("posteam")["pass"].mean()
    t["pa"] = o[o["pass"] == 1].groupby("posteam").is_play_action.mean()
    return t

def defense_metrics(season):
    """One row per defense: charted coverage on dropbacks, plus FTN blitzers and rushers."""
    c = cov[cov.season == season]
    db = snaps[(snaps.season == season) & (snaps.qb_dropback == 1)]
    t = c.groupby("defteam").agg(man=("man", "mean"), two_high=("two_high", "mean"), labelled=("man",
"size"))
    t["blitz"] = db.groupby("defteam").n_blitzers.apply(lambda s: (s.dropna() > 0).mean())
    t["rush5"] = db.groupby("defteam").number_of_pass_rushers.apply(lambda s: (s.dropna() >= 5).mean())
    return t

OFF25, DEF25 = offense_metrics(2025), defense_metrics(2025)
LG_OFF = offense_metrics(2025).mean() # average team (each team counts once)
```

⁵2026 coordinators from the course fact sheet (checked October 2026) and the reporting it cites, including <https://www.profootballrumors.com/2026/02/2026-nfl-offensive-defensive-coordinator-search-tracker> and https://en.wikipedia.org/wiki/2026_NFL_season.

```

DASH = [("proe", "PROE", lambda v: f"{v:+.1f}"), ("p11", "11", None), ("p12", "12", None),
        ("heavy", "Heavy", None), ("uc", "Under\ncenter", None), ("uc_pass", "Pass\nfrom UC", None),
        ("motion", "Motion", None), ("pa", "Play-\naction", None), ("nohuddle", "No-\nhuddle", None)]
keys = [k for k, _, _ in DASH]
D = OFF25.sort_values("proe", ascending=False)[keys]
Z = ((D - D.mean()) / D.std()).clip(-2.5, 2.5)
cmap = LinearSegmentedColormap.from_list("dev", [style.SERIES[1], style.DIVERGING[1], style.SERIES[0]])

fig, ax = plt.subplots(figsize=(6.4, 8.3))
ax.imshow(np.vstack([Z.values, np.zeros((1, len(keys)))]), cmap=cmap, vmin=-2.6, vmax=2.6, aspect="auto")
for i, tm in enumerate(list(D.index) + ["League"]):
    row = D.loc[tm] if tm != "League" else D.mean()
    for j, (k, _, fmt) in enumerate(DASH):
        z = Z.iloc[i, j] if tm != "League" else 0
        txt = fmt(row[k]) if fmt else f"{100 * row[k]:.0f}"
        ax.text(j, i, txt, ha="center", va="center", fontsize=6.9,
                color="white" if abs(z) > 1.9 else style.INK, fontweight="bold" if tm == "League" else "normal")
ax.add_patch(plt.Rectangle((-0.5, len(D) - 0.5), len(keys), 1, fc=style.GRID, ec="none", zorder=0.5))
ax.set_xticks(range(len(keys)), [lab for _, lab, _ in DASH], fontsize=7.2)
ax.xaxis.tick_top()
ax.set_yticks(range(len(D) + 1), list(D.index) + ["League"], fontsize=7)
ax.tick_params(length=0)
for sp in ax.spines.values():
    sp.set_visible(False)
ax.set_xticks(np.arange(-0.5, len(keys), 1), minor=True)
ax.set_yticks(np.arange(-0.5, len(D) + 1, 1), minor=True)
ax.grid(False)
ax.grid(which="minor", color=style.SURFACE, lw=1.6)
ax.tick_params(which="minor", length=0)
ax.axhline(len(D) - 0.5, color=style.INK_2, lw=1.2)
ax.text(-0.5, len(D) + 1.05, "PROE in percentage points; every other column is a percentage of neutral
early-down snaps.",
        fontsize=6.6, color=style.INK_2, va="top")
plt.show()

```

	PROE	11	12	Heavy	Under center	Pass from UC	Motion	Play-action	No-huddle
ARI	+8.0	40	38	19	45	41	50	43	5
LA	+5.9	55	11	34	72	46	69	48	6
CIN	+4.9	59	36	5	25	29	62	26	0
KC	+4.7	52	31	17	26	28	50	27	1
JAX	+4.2	63	30	6	47	33	65	36	6
NE	+3.9	42	30	27	52	34	54	34	2
LAC	+3.7	51	5	39	39	35	56	34	1
IND	+1.9	55	34	11	32	43	56	42	4
DEN	+1.4	60	16	22	41	29	52	35	11
SF	+0.6	24	13	63	56	40	79	38	3
PHI	-0.3	57	35	7	25	29	53	36	25
PIT	-0.4	43	35	21	43	32	62	33	8
CHI	-0.7	46	41	10	62	41	62	51	4
HOU	-0.7	65	12	17	46	36	58	36	3
DAL	-1.3	59	20	19	50	37	65	47	4
NO	-1.8	69	22	5	30	28	68	33	22
LV	-1.9	49	41	10	50	30	58	48	5
BUF	-2.2	56	15	29	56	31	66	37	4
MIN	-2.3	49	29	21	59	33	57	45	6
TEN	-3.5	57	37	6	44	31	56	42	3
TB	-3.5	61	28	10	46	27	68	32	4
CAR	-4.0	55	30	15	45	27	57	40	2
WAS	-4.1	49	40	8	17	13	66	41	61
MIA	-4.1	20	9	68	36	40	78	49	2
CLE	-4.5	38	53	9	48	31	59	44	4
SEA	-5.1	30	41	29	68	37	57	40	7
NYG	-5.1	45	49	6	33	36	40	44	26
DET	-5.4	48	39	11	65	34	65	43	4
GB	-5.7	50	42	7	45	39	64	36	5
ATL	-5.9	32	48	21	24	40	76	38	9
BAL	-11.2	23	36	42	43	27	58	36	2
NYJ	-12.2	47	34	19	42	32	74	38	4
League	-1.5	48	31	20	44	33	61	39	8

PROE in percentage points; every other column is a percentage of neutral early-down snaps.

Figure 7: The 2025 tendency dashboard: every offense on neutral first and second downs, sorted by PROE. Each cell is the team's value; its colour is how far that value sits from the league's average team (blue above, orange below, pale near average, scaled by the spread across the 32 teams). 'Heavy' = two or more backs or three or more tight ends. 'Pass from UC' = dropback rate when the quarterback was under center. Motion = FTN's is_motion (motion before or at the snap). Play-action is a share of dropbacks. Bottom row: the average team. Sources: nflverse play-by-play, participation (personnel) and FTN charting (alignment, motion, play-action, no-huddle).

Read the dashboard by columns first, then rows.

- **Heavy personnel** is the most lopsided column. Miami and San Francisco sat in a fullback or extra-tight-end package on about two early downs in three; Baltimore, the Chargers and the Rams came next, at a third or more. The 2025 shift toward heavier sets (Next Gen Stats counted fewer than three wide receivers on 41.7% of plays) shows up here as a handful of deep-blue cells in this column, not as an even drift across the league.⁶

⁶James Reber (NFL Next Gen Stats), "NFL trend watch: heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks" (offenses used fewer than three wide receivers on 41.7% of 2025 plays): <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>.

- **Under center** is led by the Rams, at 72%, and the Rams also threw from under center far more often than the league (46% against an average team's 33%). That combination is a self-scouted offense: the look that says "run" for most teams says "coin flip" for them. At the other end, Washington, Atlanta, Cincinnati, Philadelphia and Kansas City were in the shotgun or pistol on about three early downs in four or more, and Cincinnati and Kansas City also used play-action least: pass-first offenses that don't need the fake.
- **Motion** runs from the Giants (40%, about ten points below the next-lowest offense) to Miami and San Francisco near 80%, two Shanahan-tree offenses.
- **No-huddle** has one outlier: Washington, at 61% of neutral early downs, a deliberate tempo identity no other team came close to.

Then read a row as a sentence. The Rams' row says: passes more than expected, in heavy and 11 personnel, mostly from under center, with motion on most snaps and play-action on about half its dropbacks, and rarely in a hurry. That is the start of a scouting report, and the last section of this chapter turns it into one.

i Go deeper: why the motion numbers you read disagree

FTN's `is_motion` means "motion occurred on the play before **or at** the time of the snap." It is not motion *at* the snap (ESPN's series: about 4% of plays in 2017, about 22% in 2023), and it is not PFF's "shift or motion" rate (63.9% in 2025, against FTN's 55% on all 2025 runs and dropbacks).⁷ Motion and Shifts lays the three series side by side. For a tendency card the rule is simple: compare teams inside one series, and say which one.

Coverage tendencies from charting data

Offensive tendencies come mostly from things anyone can count: who is on the field, where the quarterback stands, whether somebody moved. Coverage is different. Nobody in the public data knows what the defensive coordinator called. What exists is **coverage charting**: an analyst watches each pass play after the game and labels the coverage the defense *played after the snap* (Cover 1, Cover 3, quarters and so on), plus whether it was man or zone. In nflverse these labels sit in the participation file: from the NFL's Next Gen Stats for 2018–2022 (film charting, and from 2022 a coverage model trained on that charting and fed by tracking data), and from FTN's charters for 2023 onward.⁸ They exist only on dropbacks, because coverage is something you can only see when there are routes to cover.

The phrase "after the snap" carries a lot of weight. A defense that shows one picture and plays another (the disguise of Disguise, Rotation and Split-Field) gets the label of what it *played*. Figure 8 is the classic case.

```
off = offense("gun_2x2")
nick = defense("nickel", "two_high", off)
rot = Play(off, nick, los=50, to_go=10, title="Shown two-high, played Cover 3")
rot.route("X", r.go(17)); rot.route("Z", r.go(17)); rot.route("H", r.seam(15)); rot.route("Y", r.curl(10))
rot.route("RB", r.flat(1.5, 5), side=-1)
rot.dropback(2.5)
for k in ("SDE", "SDT", "WDT", "WDE"):
    rot.rush(k)
rot.drop("FS", (15.5, 0.0), speed=6.5)      # rotates to the deep middle
rot.drop("SS", (8.0, 5.0), speed=4.5)      # comes down to the hook-curl
rot.drop("LCB", (15.0, -14.5)); rot.drop("RCB", (15.0, 14.5)) # bail to deep thirds
rot.drop("NB", (7.0, 12.5)); rot.drop("MIKE", (7.0, -3.0)); rot.drop("WILL", (6.5, -11.0))
show_animation(rot, times=[-0.5, 0.5, 1.2, 2.1])
```

⁷PFF: Daire Carragher, "NFL pre-snap motion usage reaches new highs across the NFL," February 23, 2026 (63.9% shift or motion in 2025): <https://www.pff.com/news/nfl-pre-snap-motion-usage-reaches-new-highs-across-the-nfl>. ESPN Analytics' motion-at-the-snap figures (4% of plays in 2017, 22% in 2023), from Seth Walder as quoted by Ben Solak, The Ringer, January 25, 2024: <https://www.theringer.com/nfl/2024/1/25/24050457/kansas-city-chiefs-baltimore-ravens-afc-championship-steve-spagnuolo-mike-macdonald-defense-motion>.

⁸nflverse participation data: seasons 2016–2022 from NFL Next Gen Stats, 2023 on from FTN; coverage fields filled on about 41–43% of rows in 2018–2022 and 49–50% in 2023–2025 (essentially all dropbacks), empty for 2016–2017. Course fact sheet §7 and the participation dictionary cited above. Next Gen Stats' coverage classifier (launched for 2022) uses exactly the eight pre-2023 labels (Cover 0, 1 and 2 man; Cover 2, 3, 4 and 6 zone; prevent); it reads player-tracking data and was trained on 2018–2020 plays charted by football specialists from film, scoring 88.9% on the 2021 season: AWS Machine Learning Blog, "Identifying defense coverage schemes in NFL's Next Gen Stats," February 10, 2023, <https://aws.amazon.com/blogs/machine-learning/identifying-defense-coverage-schemes-in-nfls-next-gen-stats/>. nflverse does not document, season by season, whether a pre-2023 label is a human chart or the model's.

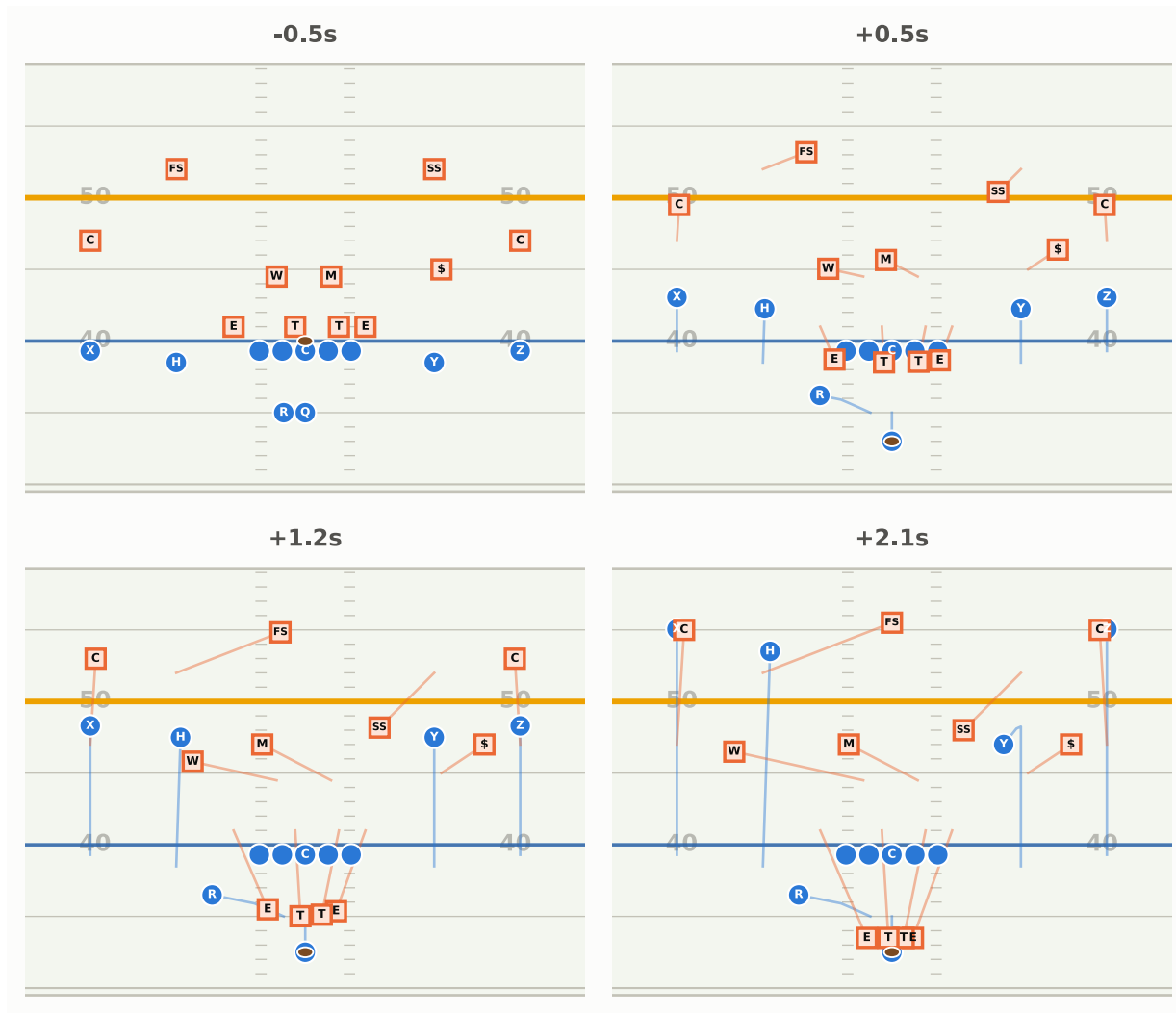


Figure 8: Shown two-high, played Cover 3 (a simulated play drawn with gridiron). Before the snap both safeties sit 12 yards deep, one on each hash side: the picture of a two-high shell. At the snap the free safety (FS) runs to the deep middle and the strong safety (SS) comes down to an underneath zone, while the corners (C) bail to deep thirds. A charter labels this snap COVER_3 and zone: the label records what the defense played, not what it showed. A count of pre-snap shells would have called it two-high.

```
COV_ORDER = ["COVER_0", "COVER_1", "2_MAN", "COVER_3", "COVER_9", "COVER_2", "COVER_4", "COVER_6"]
COV_NAME = {"COVER_0": "Cover 0", "COVER_1": "Cover 1", "2_MAN": "2-Man", "COVER_3": "Cover 3",
            "COVER_9": "Cover 9", "COVER_2": "Cover 2", "COVER_4": "Quarters", "COVER_6": "Cover 6"}
COV_COLOR = {"COVER_0": style.DEFENSE_DARK, "COVER_1": style.DEFENSE, "2_MAN": "#f5a27f",
            "COVER_3": style.SEQUENTIAL_BLUE[5], "COVER_9": style.SEQUENTIAL_BLUE[2],
            "COVER_2": "#0f7a55", "COVER_4": style.THIRD, "COVER_6": "#a6e3c8"}
```

```
def coverage_mix(c):
    return c.defense_coverage_type.value_counts(normalize=True).reindex(COV_ORDER).fillna(0)
```

```
c25 = cov[cov.season == 2025]
LG_MIX25 = coverage_mix(c25)
EARLY_MIX = coverage_mix(c25[c25.down <= 2])
THIRD_MIX = coverage_mix(c25[(c25.down == 3) & c25.ydstogo.between(4, 8)])
display(md_table(pd.DataFrame({
    "Coverage (2025 label)": [COV_NAME[k] for k in COV_ORDER],
```

```
"All dropbacks %": LG_MIX25.values * 100, "1st & 2nd down %": EARLY_MIX.values * 100,
"3rd & 4-8 %": THIRD_MIX.values * 100}), digits=1))
```

Coverage (2025 label)	All dropbacks %	1st & 2nd down %	3rd & 4-8 %
Cover 0	3.8	2.8	6.2
Cover 1	22.5	19.7	35.4
2-Man	4.7	3.7	7.4
Cover 3	27.9	30.1	18.0
Cover 9	2.6	2.8	1.8
Cover 2	23.1	24.0	21.6
Quarters	10.1	10.9	5.9
Cover 6	5.4	6.1	3.8

League-wide, 2025's labelled dropbacks were mostly Cover 3, Cover 2 and Cover 1, with quarters next. The situational split is the first tendency worth knowing: on third-and-4 to 8, man coverage (Cover 1, Cover 0, 2-Man) jumped from about 26% of early-down dropbacks to 49%. Defenses play more man when they know a pass is coming and the sticks are close: man coverage challenges every short route at once (Man Coverage). Figure 9 shows the whole league, team by team.

```
mix = c25.groupby("defteam").defense_coverage_type.value_counts(normalize=True).unstack().reindex(columns=COV_ORDER).
mix = mix.loc[mix[["COVER_0", "COVER_1", "2_MAN"]].sum(1).sort_values().index]

fig, ax = plt.subplots(figsize=(6.4, 7.0))
left = np.zeros(len(mix))
for k in COV_ORDER:
    ax.barh(range(len(mix)), mix[k] * 100, left=left, color=COV_COLOR[k], height=0.78,
            label=COV_NAME[k], edgecolor=style.SURFACE, lw=0.4)
    left += mix[k].values * 100
ax.set_yticks(range(len(mix)), mix.index, fontsize=7)
ax.set_xlim(0, 100)
ax.set_xlabel("Share of charted dropbacks (%)")
ax.grid(axis="y", visible=False)
ax.tick_params(axis="y", length=0)
ax.legend(ncol=4, fontsize=7, loc="lower center", bbox_to_anchor=(0.5, 1.0), handlelength=1.0,
          columnspacing=1.0)
ax.set_ylim(-0.6, len(mix) - 0.4)
fig.text(0.02, 0.005, "Source: FTN charting via nflverse participation data (load_participation), 2025
regular season.\n"
        "Labels exist only on dropbacks; 'blown' and 'combination' tags (fewer than 1 snap in 200) are
left out.\n"
        "FTN's labels are not comparable with the 2018-2022 Next Gen Stats labels: compare teams within
a season.",
        fontsize=6.6, color=style.INK_2, va="bottom")
fig.subplots_adjust(bottom=0.13, top=0.93, left=0.09, right=0.98)
plt.show()
```

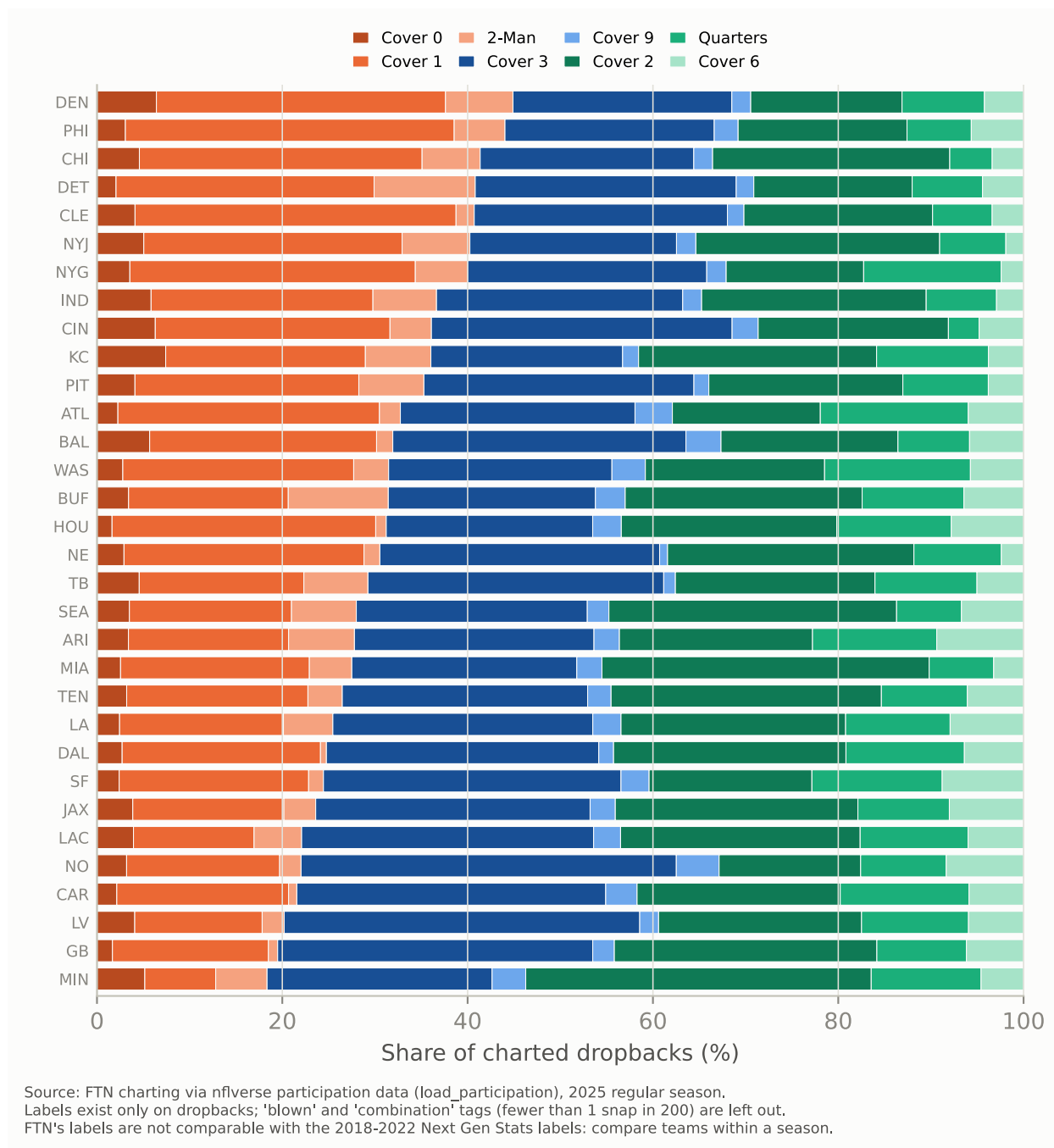


Figure 9: Coverage mix by defense, 2025 regular season: the share of each defense's charted dropbacks in each coverage label, sorted by the share played as man coverage (oranges). Blues are single-high zone (Cover 3 and Cover 9), greens two-high zone (Cover 2, quarters, Cover 6). 2-Man is a two-high man coverage, so it sits in both families. The caveat printed under the chart is part of the chart: these are one charting source's judgements for one season.

i Film room: Minnesota Vikings 2025

Brian Flores's Vikings defense (he has been Minnesota's defensive coordinator since 2023, and still is in 2026) sits at one end of Figure 9, and the dashboard row behind it is stranger than the bar.⁹ Minnesota played man coverage on just 18% of its charted 2025 dropbacks, the fewest in the league, and a two-high coverage on 59%, the most. Yet it sent at least one blitz on 51% of dropbacks, the most in the league again (the average defense: 29%).

What to notice: "blitz" and "man" usually travel together, because a defense that sends extra rushers has fewer players left to cover and the simplest way to cover with fewer is man-to-man. The Vikings broke that link: pressure from unexpected places with zone behind it, the simulated pressure and zone-blitz family of Zone Blitzes and Simulated Pressure, on top of the two-high-to-Cover-0 range 06-06 charted for 2023–2025. A tendency card that said only "blitzes a lot" would send an offense into the game expecting man coverage, which is exactly the wrong answer.

When the labels change: the 2022/2023 source break

Here is the most important caveat in the chapter, drawn. Figure 10 follows four league-wide coverage shares from 2018 to 2025, the full span for which nflverse has labels.

```
TREND = cov.groupby("season").agg(man=("man", "mean"), two_high=("two_high", "mean"),
                                c3=("defense_coverage_type", lambda s: (s == "COVER_3").mean()),
                                c1=("defense_coverage_type", lambda s: (s == "COVER_1").mean())) * 100
NGS_TWO_HIGH = {2018: 32.9, 2021: 37.8, 2024: 40.4, 2025: 42.0} # published by NFL Next Gen Stats

fig, ax = plt.subplots(figsize=(6.4, 3.6))
ax.axvspan(2022.5, 2025.5, color=style.GRID, alpha=0.55, lw=0)
ax.text(2020.0, 55.5, "Next Gen Stats labels", ha="center", fontsize=7.5, color=style.INK_2)
ax.text(2024.0, 55.5, "FTN labels", ha="center", fontsize=7.5, color=style.INK_2)
SER = [("two_high", "Two-high", style.SERIES[0]), ("man", "Man", style.SERIES[1]),
       ("c3", "Cover 3", style.SERIES[2]), ("c1", "Cover 1", style.SERIES[6])]
for k, lab, col in SER:
    ax.plot(TREND.index, TREND[k], color=col, lw=2, marker="o", ms=3.5)
    ax.text(2025.18, TREND[k].iloc[-1], lab, fontsize=7.5, color=style.INK, va="center")
ax.plot(list(NGS_TWO_HIGH), list(NGS_TWO_HIGH.values()), ls="", marker="o", ms=6.5, mfc="none",
        mec=style.INK_2, mew=1.0, label="Next Gen Stats published two-high rate")
ax.set_xlim(2017.6, 2026.1)
ax.set_ylim(15, 58)
ax.set_xticks(range(2018, 2026))
ax.set_ylabel("Share of labelled dropbacks (%)")
ax.grid(axis="x", visible=False)
ax.legend(fontsize=7, loc="lower left", frameon=False)
ax.set_title("The coverage labels changed hands after 2022")
plt.show()
```

⁹Brian Flores, Vikings defensive coordinator since 2023 and retained for 2026 (course fact sheet; <https://www.vikings.com/team/coaches-roster/brian-flores>; he joined on February 6, 2023). Coverage and blitz shares are this chapter's calculation from nflverse participation (FTN-charted coverage) and FTN charting (n_blitzers).

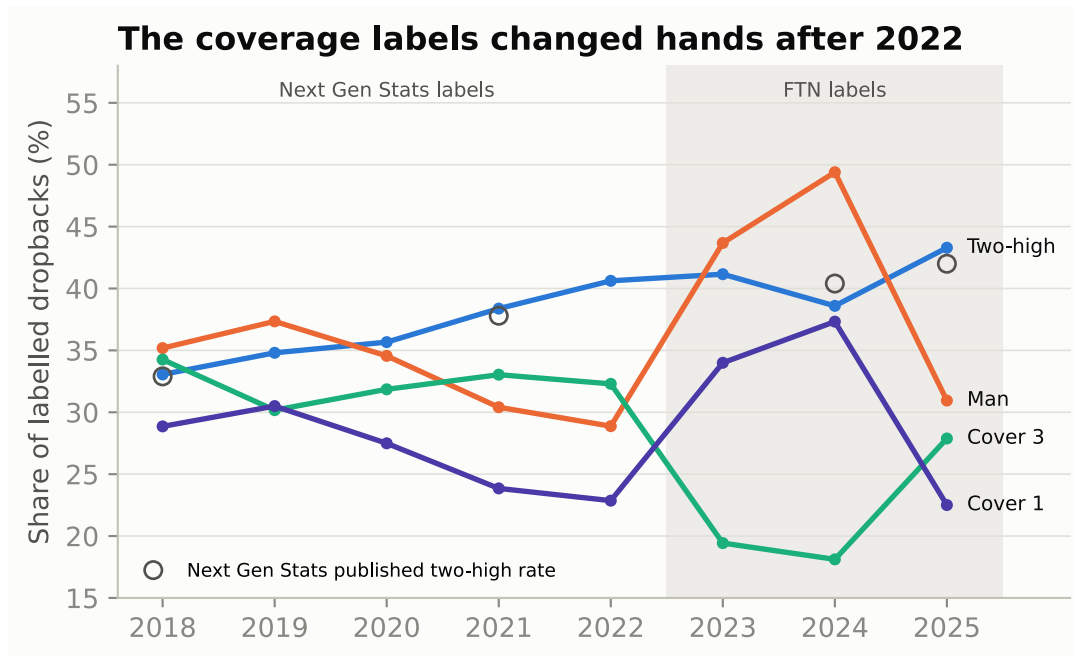


Figure 10: League-wide coverage shares on labelled regular-season dropbacks, 2018-2025, from nflverse participation data (blown, combination and prevent labels left out). The shaded seasons were charted by FTN; the earlier ones came from Next Gen Stats. Man coverage jumps from 29% to 44% at the break, to 49% in 2024, then back to 31%; Cover 3 and Cover 1 swap places and swap back. Football didn’t change that fast: the coding did. The two-high share (blue) is the one series that survives the break, and it tracks Next Gen Stats’ own published two-high rate (hollow circles) closely.

If you charted the man-coverage line without the shading, you’d write a story about a man-coverage revolution in 2023–2024 and a sudden retreat in 2025. There was no such revolution. Two different labelling operations, with different definitions and different people, met at the 2022/2023 boundary, and the FTN side itself seems to have drawn the man/zone line differently from one year to the next.¹⁰ The two-high series survives because “how many deep safeties after the snap” is easier to agree on than “was that underneath defender in man or in a zone that matches routes”: it matches the Next Gen Stats series published from tracking data (43.3% against 42.0% in 2025).

The coverage labels are not the only thing that changed at the break. In the participation file:

- **Formation labels** were SHOTGUN, SINGLEBACK, I_FORM, EMPTY, PISTOL, JUMBO and WILDCAT through 2022, and only SHOTGUN, UNDER CENTER and PISTOL from 2023. (That is why this chapter takes alignment from FTN’s `qb_location` instead.)
- **Coverage labels** gained COVER_9, COMBO and BLOWN in 2023 and lost PREVENT.
- **Personnel strings** changed format, from “1 RB, 1 TE, 3 WR” to a full list such as “1 C, 2 G, 1 QB, 1 RB, 2 T, 1 TE, 3 WR”. The `personnel_code` function above reads both.
- `ngs_air_yards` is empty from 2024; use play-by-play’s `air_yards`.

The working rule: **compare teams within a season; compare seasons only within one source, and only after checking against an independent series.** If a chart must cross the break, shade it, as Figure 10 does.

The caveats of charting data, in one place

Everything in the last three sections rests on people (or, for some fields, sensors) turning video into columns. Before you quote a charted number, run down this list.

1. **Definitions.** Every flag means exactly what its dictionary says and no more. `is_motion` is motion before *or at* the snap. `n_blitzers` counts defenders FTN judged to be blitzing, which is not the same as “five or more rushers” (`number_of_pass_rushers` in participation): a linebacker can blitz while a lineman drops into coverage, leaving four rushers. Play-action is a run fake, but how convincing a fake has to be is the charter’s call.

¹⁰Man share of labelled dropbacks in nflverse participation: 28.8% (2022), 41.2% (2023), 48.3% (2024), 30.8% (2025), all blown/combination labels included (course fact sheet §7, which flags the jump as a coding artefact); this chapter’s figure leaves those labels out. Next Gen Stats’ published two-high rate: 32.9% (2018), 37.8% (2021), 40.4% (2024), 42.0% (2025), <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm>.

2. **Judgement and error.** Coverage is inferred, from video by someone who doesn't know the call or, for the later Next Gen Stats labels, by a model trained on such charting (about 89% right on the eight labels in its makers' own test). Match coverages, which start as zone and turn into man by rule (pattern matching), and late rotations make "Cover 3 or Cover 1?" a genuine judgement. FTN's COMBO and BLOWN tags exist because some snaps don't fit a single label. No public figure tells you how often two charters would agree on the same play.
3. **What isn't labelled.** Coverage exists only on dropbacks, so a defense's coverage mix is its mix *against passes*, which partly reflects which downs and distances opponents chose to throw on. Personnel and box counts fill every row of the participation file from 2023 on but about three in four before; formation labels fill about three rows in four before 2023 and four in five after.
4. **The source break** (above): labels, formations and personnel strings all change between the 2022 and 2023 seasons.
5. **Timing.** FTN charting arrives game by game during the season, but participation (personnel and coverage) arrives only after the season ends, so the current season's personnel and coverage are missing until about February.
6. **Sample size.** A team's season has four or five hundred neutral early downs and about six hundred dropbacks, but a cell like "coverage on third-and-medium" holds perhaps fifty or sixty. A share's standard error is about $\sqrt{p(1-p)/n}$: at 30% from sixty snaps, plus or minus six points.
7. **Licence and credit.** FTN's charting is free to use under CC-BY-SA 4.0, which means you must credit FTN, and anything you publish from it must carry the same licence.

Common misconception

"The coverage label is what the defense called." It's what one charter, watching afterwards, judged the defense to have played. Usually that's close to the call. But a well-disguised snap, a match coverage or a blown assignment can be labelled differently from what the coordinator signalled, and a different charting company might label the same snap differently again. Treat a team's coverage mix as strong evidence about its habits and weak evidence about any single snap.

A one-page scouting report

Everything so far comes together in one artefact: a card you could hand a coach on Friday or carry to a game on Sunday. Game Planning Week showed what an NFL quality-control coach's tendency report looks like, pages of down-and-distance tables built from film. The public-data version fits on one page, and its job is narrower: tell a watcher, in a glance, **what this team does that the league doesn't**.

The example is the 2025 Rams, scouted the way Carolina's staff had to in January 2026: the Panthers, the NFC's fourth seed, hosted the Rams in the wild-card round, and the Rams won 34–31.¹¹ The card uses only regular-season data that existed before that game.

```
def stat_tile(ax, x, y, label, value, league, fmt="{:.0%}"):
    """One number, its label, and the league's number underneath (axes coordinates 0-1)."""
    ax.text(x, y, fmt.format(value), fontsize=13, fontweight="bold", color=style.INK, ha="left",
            va="baseline", transform=ax.transAxes)
    ax.text(x, y - 0.17, label, fontsize=7, color=style.INK, ha="left", va="baseline", transform=ax.transAxes)
    ax.text(x, y - 0.31, "league " + fmt.format(league), fontsize=6.5, color=style.INK_2, ha="left",
            va="baseline", transform=ax.transAxes)

def panel(fig, rect, title, title_x):
    """An axes at `rect`, with its title set flush left at figure x = title_x."""
    ax = fig.add_axes(rect)
    fig.text(title_x, rect[1] + rect[3] + 0.018, title, fontsize=8.6, color=style.INK, fontweight="bold",
            va="bottom")
    return ax

def dot_rows(ax, rows):
```

¹¹2025–26 NFL playoffs, wild-card round, January 10, 2026, Bank of America Stadium: Los Angeles Rams 34, Carolina Panthers 31 (Stafford's 19-yard touchdown pass to Colby Parkinson with 38 seconds left); Carolina was the NFC's fourth seed (8–9, NFC South champion), the Rams the fifth (12–5). Box score: <https://www.pro-football-reference.com/boxscores/202601100car.htm>; seeds: https://en.wikipedia.org/wiki/2025%E2%80%9326_NFL_playoffs.

```

"""rows: (label, team pass rate or None, league pass rate). Team = blue dot, league = grey tick."""
yy = np.arange(len(rows))[:-1]
for yv, (_, team_v, lg_v) in zip(yy, rows):
    if team_v is not None:
        ax.scatter(team_v * 100, yv, s=40, color=style.SERIES[0], zorder=3)
        ax.scatter(lg_v * 100, yv, s=60, marker="|", color=style.MUTED, lw=2, zorder=2)
ax.set_yticks(yy, [r[0] for r in rows], fontsize=6.6)
ax.set_xlim(0, 100)
ax.set_ylim(-0.6, len(rows) - 0.4)
ax.axvline(50, color=style.BASELINE, lw=0.8)
ax.tick_params(axis="x", labelsize=6.5)
ax.grid(axis="y", visible=False)

TELLS = { # metric: (offense or defense table, phrase when high, phrase when low)
    "proe": ("off", "passes more than the situation predicts", "runs more than the situation predicts"),
    "heavy": ("off", "lives in heavy personnel (2+ backs or 3+ TEs)", "rarely uses heavy personnel"),
    "uc": ("off", "plays under center a lot", "lives in the shotgun"),
    "uc_pass": ("off", "throws from under center (the alignment tell is broken)", "runs from under center"),
    "motion": ("off", "moves before the snap on most plays", "rarely uses motion"),
    "pa": ("off", "fakes the run on many dropbacks", "rarely uses play-action"),
    "nohuddle": ("off", "plays without a huddle", "huddles almost every snap"),
    "man": ("def", "plays a lot of man coverage", "plays zone behind almost everything"),
    "two_high": ("def", "plays two-high after the snap", "plays single-high after the snap"),
    "blitz": ("def", "blitzes often", "rarely blitzes"),
}

def biggest_tells(team, off_m, def_m, n=3):
    """The n measures on which the team sits furthest from the average team, in standard deviations."""
    rows = []
    for k, (side, hi, lo) in TELLs.items():
        t = off_m if side == "off" else def_m
        z = (t.loc[team, k] - t[k].mean()) / t[k].std()
        rows.append((abs(z), z, k, hi if z > 0 else lo, t.loc[team, k], t[k].mean()))
    return sorted(rows, reverse=True)[:n]

def scouting_card(team, season, name=None):
    """A one-page tendency card for one team-season, built from public nflverse data."""
    off_m, def_m = offense_metrics(season), defense_metrics(season)
    lg = snaps[(snaps.season == season) & snaps.early_neutral]
    ot = lg[lg.posteam == team]
    fig = plt.figure(figsize=(6.6, 7.7))
    fig.text(0.02, 0.99, f"OPPONENT CARD {name or team}, {season} regular season", fontsize=12,
            fontweight="bold", color=style.INK, va="top")
    fig.text(0.02, 0.957, "Offense: neutral 1st & 2nd downs (WP 20-80%, outside the last 2:00 of each
half), "
            f"{len(ot)} snaps. Defense: charted dropbacks.\nBlue = {team}, grey = league. "
            "Sources: nflverse play-by-play, participation, FTN charting.", fontsize=6.8,
            color=style.INK_2, va="top", linespacing=1.3)
    L, R = (0.19, 0.29), (0.70, 0.28) # left and right columns: (x, width) of the axes

    # A. PROE by situation
    ax = panel(fig, [L[0], 0.585, L[1], 0.29], "Run or pass? PROE by situation", 0.02)

```

```

ps = proe_by_situation(team, season)
yy = np.arange(len(ps))[:-1]
ax.barh(yy, ps.proe, color=[style.SERIES[0] if v > 0 else style.SERIES[1] for v in ps.proe], height=0.62)
ax.axvline(0, color=style.INK_2, lw=0.9)
ax.set_yticks(yy, [f"{s} ({n})" for s, n in zip(ps.index, ps.n)], fontsize=6.6)
lim = max(20, np.ceil(ps.proe.abs().max() / 10) * 10)
ax.set_xlim(-lim, lim)
ax.tick_params(axis="x", labelsz=6.5)
ax.set_xlabel("points over expected (snaps)", fontsize=6.6)
ax.grid(axis="y", visible=False)

# B. personnel: how often, and how often they pass from it
ax = panel(fig, [R[0], 0.585, R[1], 0.29], "Personnel: pass rate from each", 0.52)
share = ot.pers.value_counts(normalize=True)
rows = []
for g in ["11", "12", "13", "21", "22"]:
    tg = ot[ot.pers == g]
    rows.append((f"{g}: {share.get(g, 0):.0%} of snaps", tg["pass"].mean() if len(tg) >= 10 else None,
                lg[lg.pers == g]["pass"].mean()))
dot_rows(ax, rows)
ax.set_xlabel("pass rate (%): team dot, league tick", fontsize=6.6)

# C. where the quarterback stands
ax = panel(fig, [L[0], 0.375, L[1], 0.12], "Where the QB stands: pass rate", 0.02)
rows = []
for code, lab in [("U", "Under center"), ("S", "Shotgun"), ("P", "Pistol")]:
    tg = ot[ot.qb_location == code]
    rows.append((f"{lab}: {(ot.qb_location == code).mean():.0%}", tg["pass"].mean() if len(tg) >= 10
else None,
                lg[lg.qb_location == code]["pass"].mean()))
dot_rows(ax, rows)

# D. offensive tiles
ax = panel(fig, [0.53, 0.355, 0.45, 0.14], "Pre-snap habits (play-action: share of dropbacks)", 0.52)
ax.axis("off")
for i, (k, lab) in enumerate([("motion", "motion"), ("pa", "play-action"), ("nohuddle", "no-huddle"),
                             ("proe", "PROE (points)"))]:
    fmt = "{:+.1f}" if k == "proe" else "{:.0%}"
    stat_tile(ax, i * 0.26, 0.62, lab, off_m.loc[team, k], off_m[k].mean(), fmt)

# E. coverage after the snap
ax = panel(fig, [0.24, 0.10, 0.34, 0.18], "Coverage after the snap (charted dropbacks)", 0.02)
c = cov[cov.season == season]
ct = c[c.defteam == team]
bars = [(f"{team}, 1st-2nd down", coverage_mix(ct[ct.down <= 2]), len(ct[ct.down <= 2])),
        ("League", coverage_mix(c[c.down <= 2]), None),
        (f"{team}, 3rd down", coverage_mix(ct[ct.down == 3]), len(ct[ct.down == 3])),
        ("League", coverage_mix(c[c.down == 3]), None)]
yy = np.array([3.2, 2.4, 1.0, 0.2])
for yv, (lab, mixv, n) in zip(yy, bars):
    lft = 0
    for k in COV_ORDER:
        ax.barh(yv, mixv[k] * 100, left=lft, color=COV_COLOR[k], height=0.62, edgecolor=style.SURFACE,
lw=0.4,
                alpha=1.0 if n else 0.55)

```

```

        if mixv[k] >= 0.12:
            ax.text(lft + mixv[k] * 50, yv, COV_NAME[k].replace("Quarters", "Qtrs"), fontsize=5.8,
                    ha="center", va="center",
                    color="white" if (n and k in ("COVER_0", "COVER_1", "COVER_3", "COVER_2"))
        else style.INK)
            lft += mixv[k] * 100
    ax.set_yticks(yy, [lab + (f" ({n})" if n else "") for lab, _, n in bars], fontsize=6.6)
    ax.set_xlim(0, 100)
    ax.tick_params(axis="x", labels=6.5)
    ax.grid(axis="y", visible=False)

    # F. defensive tiles
    ax = panel(fig, [0.66, 0.08, 0.32, 0.20], "Defense in four numbers", 0.62)
    ax.axis("off")
    for i, (k, lab) in enumerate([("man", "man coverage"), ("two_high", "two-high"),
                                  ("blitz", "1+ blitzer"), ("rush5", "5+ rushers")]):
        stat_tile(ax, (i % 2) * 0.5, 0.78 - (i // 2) * 0.52, lab, def_m.loc[team, k], def_m[k].mean())

    # G. the tells
    fig.text(0.02, 0.040, "Biggest tells (furthest from the average team):", fontsize=7.6,
             fontweight="bold", color=style.INK, va="top")
    for i, (_, z, k, phrase, v, m) in enumerate(biggest_tells(team, off_m, def_m)):
        fmt = (lambda q: f"{q:+.1f}") if k == "proe" else (lambda q: f"{100 * q:.0f}%")
        fig.text(0.04, 0.018 - i * 0.019, f"{i + 1}. {phrase}: {fmt(v)} (league {fmt(m)})",
                 fontsize=7.2, color=style.INK, va="top")
    return fig

```

```

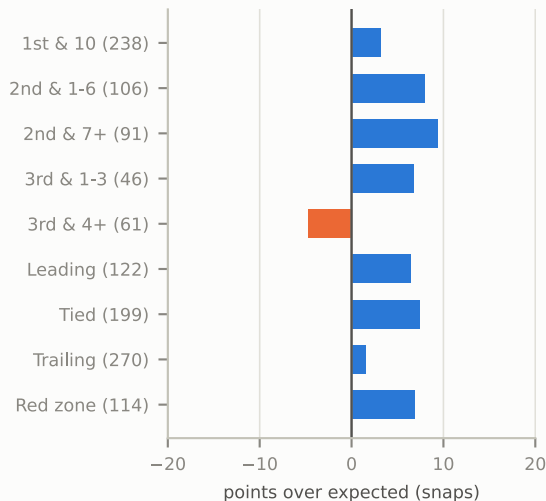
scouting_card("LA", 2025, name="Los Angeles Rams")
plt.show()

```

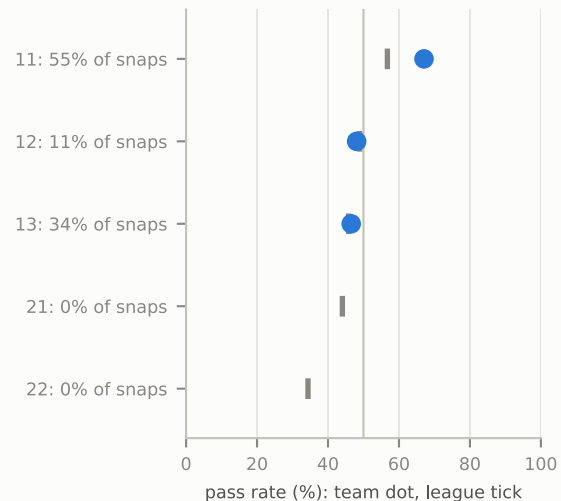
OPPONENT CARD Los Angeles Rams, 2025 regular season

Offense: neutral 1st & 2nd downs (WP 20-80%, outside the last 2:00 of each half), 467 snaps. Defense: charted dropbacks. Blue = LA, grey = league. Sources: nflverse play-by-play, participation, FTN charting.

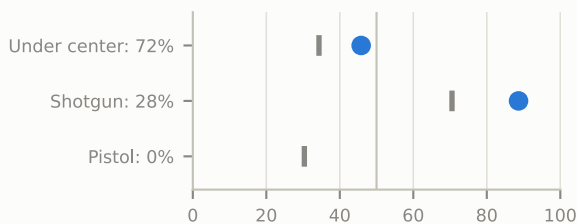
Run or pass? PROE by situation



Personnel: pass rate from each



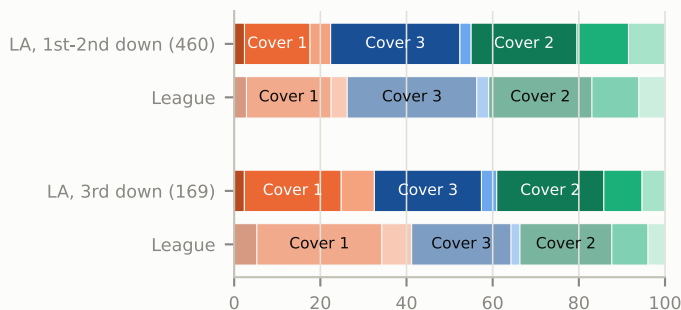
Where the QB stands: pass rate



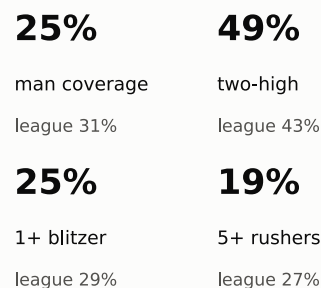
Pre-snap habits (play-action: share of dropbacks)



Coverage after the snap (charted dropbacks)



Defense in four numbers



Biggest tells (furthest from the average team):

1. plays under center a lot: 72% (league 44%)
2. throws from under center (the alignment tell is broken): 46% (league 33%)
3. passes more than the situation predicts: +5.9 (league -1.5)

Figure 11: A one-page scouting report, built entirely from public data: the 2025 Rams, as Carolina could have scouted them before the January 2026 wild-card game. Top left: PROE by situation (blue bars, more passing than expected; orange, more running), with the number of snaps behind each bar. Top right and middle left: pass rate from each personnel group and quarterback alignment, the Rams' dot against the league's tick, with how often the Rams used each. Middle right: pre-snap habits against the league. Bottom: the defense's charted coverage on early and third downs against the league, and four defensive rates. The three 'tells' at the foot are chosen automatically: the measures on which the Rams sat furthest from the average team.

```

en25 = snaps[(snaps.season == 2025) & snaps.early_neutral]
la25 = en25[en25.posteam == "LA"]
CARD = {
    "la_13_share": (la25.pers == "13").mean(), "lg_13_share": (en25.pers == "13").mean(),
    "la_13_pass": la25[la25.pers == "13"]["pass"].mean(), "lg_13_pass": en25[en25.pers == "13"]
    ["pass"].mean(),
    "la_11_pass": la25[la25.pers == "11"]["pass"].mean(), "lg_11_pass": en25[en25.pers == "11"]
    ["pass"].mean(),
    "la_uc_pass": la25[la25.qb_location == "U"]["pass"].mean(),
    "lg_uc_pass": en25[en25.qb_location == "U"]["pass"].mean(),
}
D25m = DEF25.mean()

```

Walk the card the way a defensive coach would, top to bottom.

The run/pass question. The PROE bars say the Rams passed more than expected in most situations, and the personnel panel says from what. They threw on 67% of their 11-personnel snaps (league 57%), and they lived in 13 personnel, three tight ends, a grouping the league used on only 6% of neutral early downs. The Rams used it on 34%, and passed from it about as often as the league did (47% against 46%), so the grouping itself gave nothing away; what it did was drag a defense into its bigger personnel. The quarterback-alignment panel is the real alarm. League-wide, under center meant a pass only 34% of the time; for the Rams it meant a pass 46% of the time. A defense that keyed the alignment, the most reliable tell in public data, would have been wrong far more often against the Rams than against almost anyone else.

The pre-snap habits. Motion on most snaps, play-action on a large share of dropbacks, almost no no-huddle. That is a team that will give a defense time to substitute and set, then make it adjust anyway.

The defense. The coverage bars and the four tiles describe a zone defense: man coverage on 25% of charted dropbacks (average team 31%), two-high after the snap on 49% (average 43%), and a blitz on 25% (average 29%). A little more zone and a little more two-high than the average defense, a little less blitzing; no single extreme to build a plan around, which is itself worth knowing.

What the card can't say. The public files don't record formation details (receiver splits, bunch or nub, the back's offset), the hash, the field zone in fine grain, or route concepts, all of which a pro tendency report is built on. The Rams' 2025 offense also had a play-caller (Sean McVay) and quarterback who would both be on the field for the game, so the prior was a strong one; for a 2026 opponent with a new coordinator, the card would have to lean on the 2026 games so far. And the defensive panel uses labels that only arrived after the season, so a mid-season version would have had to borrow them from the previous year, or chart them by hand.

Try it on any team-season from 2022 to 2025:

```

scouting_card("CAR", 2025, name="Carolina Panthers") # or any team abbreviation and season, 2022-2025
plt.show()

```

Grading the card

A card is a forecast, so score it like one. During the game, before each snap, write your call: run or pass, and for dropbacks, man or zone. Afterwards, load the game's play-by-play, count your hits on neutral early downs, and compare with two baselines: the team's season pass rate ("always guess the more common one") and the alignment rule from 08-02 (shotgun means pass, under center means run). If you can't beat the alignment rule, your card isn't adding anything yet; if you beat it by five points, you're reading the game better than the most reliable tell in public data. Film Study and Charting turns this into a full charting routine.

💡 Watch for it

The opponent card, live. Before the game:

- **Build the card** for the team you care about most, with `scouting_card`, from the most recent season with the same play-caller. If he is new, note that and put more weight on this season's games so far.
- **Circle three tells:** the card's own top three, or the cells furthest from league on the dashboard. Write each as a prediction ("under center: expect a pass about 45% of the time, not the league's 35%").

During the game:

- **On every early down in a close game, call run or pass before the snap**, using personnel first (count the tight ends and backs as they jog on), then the quarterback's alignment, then your card's adjustments for this team. Note when motion happens and whether it changed your call.
- **On every dropback, call the coverage after the snap:** one deep safety or two? Corners turning and running (zone) or turning with receivers (man)? Compare against the card's coverage bars.
- **Mark the moments the filter excludes:** two-minute drills and blowouts. Don't score those; the card wasn't built for them.

Afterwards, grade yourself against the two baselines above, and note which tell held and which one the opponent broke, maybe on purpose.

Predict the play

The pictures use the course's position letters: on offense X, Z and H are wide receivers, Y and U tight ends, R the running back; on defense E is an end, T a tackle, M and W linebackers, \$ the nickel back, C a cornerback, FS and SS the safeties.

💡 Predict the play

First-and-10 at your own 35, second quarter, tied. The 2025 Rams line up in 11 personnel with Stafford under center and the back behind him. League-wide, what does this look usually mean? And what should the Rams' card make you predict instead?

```
off11 = offense("singleback")
d1 = Play(off11, defense("nickel", "single_high", off11), los=45, to_go=10, title="1st & 10, own 35, tied")
d1.highlight("QB")
d1.draw(lateral=(-19, 19), window=(-10, 16), width_in=5.6)
plt.show()
```

1st & 10, own 35, tied

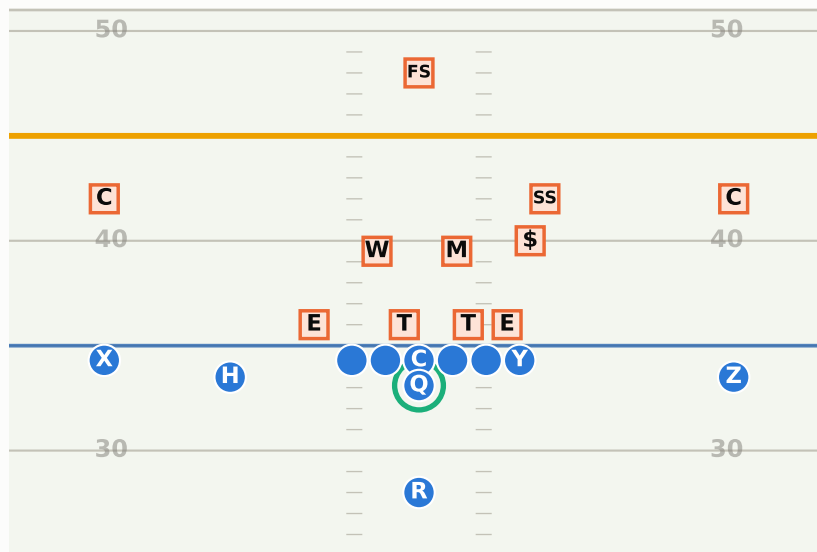


Figure 12: First-and-10 at the offense's own 35: 11 personnel (one back, one tight end, three receivers) with the quarterback under center, against a nickel defense with one deep safety. For most offenses this look means run about two times in three.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Third-and-6 at midfield. The 2025 Vikings show two deep safeties, with both inside linebackers walked up into the A gaps on either side of the center, six defenders on or near the line. You're on offense. Is pressure coming, and if it is, should the quarterback expect man coverage behind it? How far would you trust the card here?

```
off2 = offense("gun_2x2")
nick2 = defense("nickel", "two_high", off2)
spots = {"SDE": (1.0, 7.0), "SDT": (1.0, 3.6), "WDT": (1.0, -3.6), "WDE": (1.0, -7.0),
         "MIKE": (1.8, 1.25), "WILL": (1.8, -1.25)}
nick2 = [p.moved(d=spots[p.key][0], w=spots[p.key][1]) if p.key in spots else p for p in nick2]
d2 = Play(off2, nick2, los=60, to_go=6, title="3rd & 6, midfield")
d2.highlight("MIKE", "WILL")
d2.draw(lateral=(-19, 19), window=(-8, 16), width_in=5.6)
plt.show()
```

3rd & 6, midfield

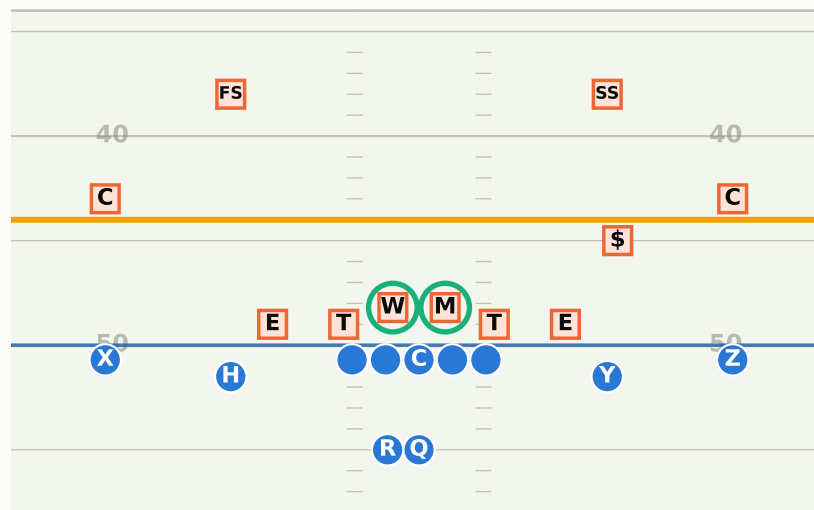


Figure 13: Third-and-6 at midfield: 11 personnel in the shotgun against a nickel defense showing a two-high shell, with both inside linebackers (M and W) mugging the A gaps beside the center. Six defenders threaten to rush; any of them may drop.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Fourth quarter, 6:10 left, down 10. First-and-10 at your own 30. The offense is in 11 personnel in the shotgun, and the play is a handoff for 4 yards. Three questions: roughly what is this snap's xpass? What does the run add to the team's `pass_oe`? And does it belong in the team's PROE?

```
off3 = offense("gun_2x2")
d3 = Play(off3, defense("nickel", "two_high", off3), los=40, to_go=10, title="Q4 6:10, down 10, 1st & 10 own 30")
d3.handoff("RB", t=0.6)
d3.run("RB", [(1.0, 1.6), (5.0, 2.6), (9.0, 3.0)], delay=0.3)
d3.highlight("RB")
d3.draw(lateral=(-19, 19), window=(-8, 16), width_in=5.6)
plt.show()
```

Q4 6:10, down 10, 1st & 10 own 30

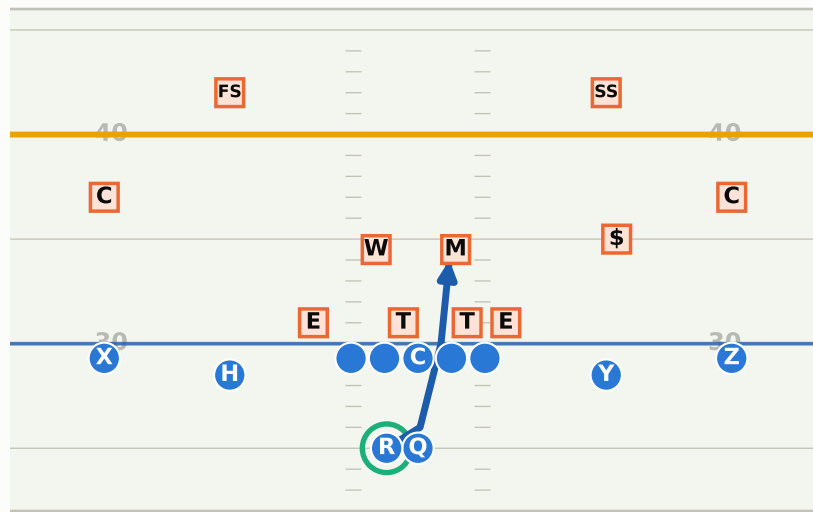


Figure 14: First-and-10 at the offense's own 30 with 6:10 left in the fourth quarter, down 10: 11 personnel in the shotgun against a nickel defense playing two deep safeties, conceding the short run to keep everything in front.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **A raw pass rate mostly measures the scoreboard.** Teams that trail pass, teams that lead run; in 2025 the correlation between a team's pass rate and its average win probability was -0.45 . Filter to **neutral situations** (win probability 20–80%, outside the last two minutes of each half, first and second down) before you read anything into it.
- **xpass is a model of the league-average play-caller**, and **PROE is the difference** between what a team did and what that model expected, in percentage points. A thirty-line logistic regression on down, distance, field position, score and clock gets close to nflverse's model.
- **PROE describes style, not quality.** It correlates only loosely with EPA (about 0.3), the arrow runs both ways, and a dominant run game (the 2024 Eagles) can drive it far below zero on the way to a Super Bowl.
- **Style is steadier than results.** Four games of PROE predict a team's rest of season better than eight games of EPA, and PROE carries over between seasons when the head coach stays, much less when he leaves.
- **Personnel, alignment, motion, play-action and no-huddle** each get one number per team; read them side by side and look for the broken tells (the Rams throwing from under center).
- **Coverage labels are charted judgements:** post-snap only, dropbacks only, and coded differently on each side of the 2022/2023 break. Compare teams within a season; never chart a cross-break trend without marking it. A one-page card puts all of it together, and grading it live tells you whether it works.

Connections

Builds on: EPA and the signal-versus-noise habit from Expected Points, EPA, and Success Rate; tendencies, the tendency table and the self-scout from Constraint Theory and Sequencing; the pro tendency report from Game Planning Week; the data files from Working with nflverse Data in Python; and the coverage vocabulary of Part 6, especially Disguise, Rotation and Split-Field.

Sets up:

- Tracking Data and the Big Data Bowl: measuring coverage and motion from player coordinates instead of charters' labels, and the limits of each.
- Scheme and Fantasy: PROE, personnel and play-action as the context behind a player's usage.
- Watching Live and Film Study and Charting: the opponent card as part of the live checklist, and charting a game yourself when the public files run out.

If you want to go deeper: the nflfastR documentation for `add_xpass()` and the nflverse data dictionaries for play-by-play, participation and FTN charting, which define every column used here;¹² Ben Baldwin's Open Source Football write-ups of the nflfastR models; and FTN's own notes on how its charting is done, before you lean hard on any single label.

Answers to Predict the play

Answer 1.

```
u11 = snaps[(snaps.season == 2025) & snaps.early_neutral & (snaps.pers == "11") & (snaps.qb_location == "U")]
la11 = u11[u11.posteam == "LA"]
LG11, RAMS11, RAMS11_N = u11["pass"].mean(), la11["pass"].mean(), len(la11)
RAMS11_PA = la11[la11["pass"] == 1].is_play_action.mean()
LG11_PA = u11[u11["pass"] == 1].is_play_action.mean()
print(f"11 personnel, under center, neutral early downs 2025: league pass rate {LG11:.0%}; "
      f"Rams {RAMS11:.0%} on {RAMS11_N} snaps")
print(f"Of those dropbacks, play-action: Rams {RAMS11_PA:.0%}, league {LG11_PA:.0%}")
```

```
11 personnel, under center, neutral early downs 2025: league pass rate 33%; Rams 54% on 151 snaps
Of those dropbacks, play-action: Rams 79%, league 84%
```

League-wide, 11 personnel under center is a run look: offenses passed on about 33% of those snaps. The Rams passed on 54% of 151: a coin flip, with a lean to the pass. When they threw from it, most of the dropbacks (79%) began with a run fake, as nearly everyone's under-center passes do (league 84%); the Rams' difference was how *often* they threw from the look, not how. So the card's prediction is not "run"; it is "either, and if it's a pass, expect a run fake first." For a defense that's a real problem: the linebackers must play the run (they can't ignore a 50% run look) and the safety must not bite on the fake. That is what a broken tell costs an opponent, and it's why the Rams built it (the tendency breaker of 08-02, turned into a whole identity).

Answer 2.

```
t3 = snaps[(snaps.season == 2025) & (snaps.qb_dropback == 1) & (snaps.down == 3) & snaps.ydstogo.between(4,
8)]
c3 = cov[(cov.season == 2025) & (cov.down == 3) & cov.ydstogo.between(4, 8)]
MIN_BLITZ3 = (t3[t3.defteam == "MIN"].n_blitzers.dropna() > 0).mean()
LG_BLITZ3 = (t3.n_blitzers.dropna() > 0).mean()
MIN_MAN3, MIN_N3 = c3[c3.defteam == "MIN"].man.mean(), len(c3[c3.defteam == "MIN"])
LG_MAN3 = c3.man.mean()
SE_MAN3 = np.sqrt(MIN_MAN3 * (1 - MIN_MAN3) / MIN_N3)
print(f"3rd & 4-8 dropbacks, 2025: Vikings sent a blitz on {MIN_BLITZ3:.0%} (league {LG_BLITZ3:.0%})")
print(f"Man coverage: Vikings {MIN_MAN3:.0%} of {MIN_N3} labelled snaps (+/- {SE_MAN3:.0%}); league {LG_MAN3:.0%}")
```

¹²nflverse data dictionaries: play-by-play https://nflreadr.nflverse.com/articles/dictionary_pbp.html, participation and FTN charting as cited above; nflreadpy: <https://github.com/nflverse/nflreadpy>; Open Source Football, including Ben Baldwin's post on the nflfastR models (which fastmodels cites for the xpass model's code): <https://opensourcefootball.com/posts/2020-09-28-nflfastR-ep-wp-and-cp-models/>.

```
assert MIN_MAN3 < LG_MAN3 and abs(MIN_BLITZ3 - LG_BLITZ3) < 0.04 and DEF25.loc["MIN", "blitz"] > MIN_BLITZ3 + 0.08
```

3rd & 4-8 dropbacks, 2025: Vikings sent a blitz on 39% (league 39%)
Man coverage: Vikings 43% of 56 labelled snaps (+/- 7%); league 49%

This drill has a twist, and it is the reason to split a card by situation. The Vikings' season number says "blitz": a blitz on 51% of all their 2025 dropbacks, the most in the league. But on third-and-4 to 8 they sent one on 39%, the same as the league (39%). Their extra pressure came on early downs, when offenses least expect it; on third-and-medium, when everybody expects pressure, they were ordinary. So the double A-gap mug here is at least as likely to be a *simulated* pressure, with one or both linebackers dropping out, as a real blitz.

Coverage tells the same story. The league played man on 49% of these snaps; the Vikings on 43% of 56 labelled snaps. So the honest prediction is "a pressure look, probably four or five rushers, probably zone behind it": throw quickly to a spot a dropping linebacker can't reach in time, and don't count on beating a man-to-man defender.

And the trust question: 56 snaps means a standard error of about 7% on that man share. The card tells you the Vikings' *style* with some confidence; on any one third down it can only shade the odds.

Answer 3.

```
XP_SCRIPT = xpass_model(xp_features(situation(1, 10, 70, -10, 370, 4)))[0]
similar = plays[(plays.qtr == 4) & plays.score_differential.between(-11, -9)
                & plays.game_seconds_remaining.between(300, 450) & (plays.down == 1)]
XP_SIMILAR = similar.xpass.mean()
print(f"Our model's xpass: {XP_SCRIPT:.2f}; this run's pass_oe on our model: {100 * (0 - XP_SCRIPT):+.0f}")
print(f"nflverse xpass on the real snaps like it: {XP_SIMILAR:.2f}")
print(f"Real first downs like it (Q4, 5:00-7:30 left, down 9-11): pass rate {similar['pass'].mean():.0%}, "
      f"average win probability {similar.wp.mean():.0%} ({len(similar):,} snaps, 2016-2025)")
```

```
Our model's xpass: 0.67; this run's pass_oe on our model: -67
nflverse xpass on the real snaps like it: 0.84
Real first downs like it (Q4, 5:00-7:30 left, down 9-11): pass rate 84%, average win probability 11% (510
snaps, 2016-2025)
```

Real offenses in this spot passed on 84% of their first downs, and nflverse's model agrees (xpass about 0.84 on those snaps). This chapter's small model says only 0.67: it is weakest in exactly these late, lopsided corners, where it has fewest examples and its simple score-and-clock terms bend least. Either way, the run's `pass_oe` is a big negative on one snap, about -84 on nflverse's model. But it doesn't belong in a *neutral* PROE: the win probability on snaps like this averages about 11%, below the 20% floor, and the filter drops it.

Why drop it, if xpass already priced the situation? Because the run might be a sensible choice that xpass can't see (the defense playing two deep safeties and inviting it, as in the picture; a team that needs two scores and wants to use clock between them), and because one snap like this moves an average more than ten neutral snaps do. Keep it, and a trailing team's coach looks more run-minded than he is; that is the Jets' raw pass rate problem from the opening, in reverse.